D4.3 Climate change

D4 Continuity and change
Ecosystems

SL

Understandings:

- D4.3.1—Anthropogenic causes of climate change
- D4.3.2—Positive feedback cycles in global warming
- D4.3.3—Change from net carbon accumulation to net loss in boreal forests as an example of a tipping point
- D4.3.4—Melting of landfast ice and sea ice as examples of polar habitat change
- D4.3.5—Changes in ocean currents altering the timing and extent of nutrient upwelling
- D4.3.6—Poleward and upslope range shifts of temperate species
- D4.3.7—Threats to coral reefs as an example of potential ecosystem collapse
- D4.3.8—Afforestation, forest regeneration and restoration of peat-forming wetlands as approaches to carbon sequestration

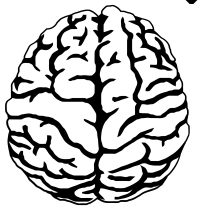
Key words:
Anthropogenic
Albedo
Greenhouse effect
Methane
Nitrous oxide
Ice cores
Fossil fuels
Deforestation
Permafrost
Peat
Boreal forest
Tipping point
Landfast ice
Nutrient upwelling
Temperate
Coral bleaching
Afforestation
Forest restoration
Forest regeneration
Sequestration

IB Learner Profile

Inquirers - We nurture our curiosity, developing skills for inquiry and research. We know how to learn independently and with others. We learn with enthusiasm and sustain our love of learning throughout life.



Knowledgeable - We develop and use conceptual understanding, exploring knowledge across a range of disciplines. We engage with issues and ideas that have local and global significance.



Thinkers - We use critical and creative thinking skills to analyse and take responsible action on complex problems. We exercise initiative in making reasoned, ethical decisions



Communicators - We express ourselves confidently and creatively in more than one language and in many ways. We collaborate effectively, listening carefully to the perspectives of other individuals and groups.



Principled - We act with integrity and honesty, with a strong sense of fairness and justice, and with respect for the dignity and rights of people everywhere. We take responsibility for our actions and their consequences.



Open minded - We critically appreciate our own cultures and personal histories, as well as the values and traditions of others. We seek and evaluate a range of points of view, and we are willing to grow from the experience.



Caring - We show empathy, compassion and respect. We have a commitment to service, and we act to make a positive difference in the lives of others and in the world around us.



Risk takers - We approach uncertainty with forethought and determination; we work independently and cooperatively to explore new ideas and innovative strategies. We are resourceful and resilient in the face of challenges and change.



Balanced - We understand the importance of balancing different aspects of our lives—intellectual, physical, and emotional—to achieve well-being for ourselves and others. We recognize our interdependence with other people and with the world in which we live.



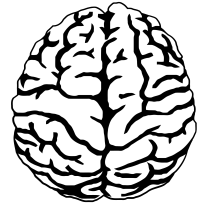
Reflective - We thoughtfully consider the world and our own ideas and experience. We work to understand our strengths and weaknesses in order to support our learning and personal development.



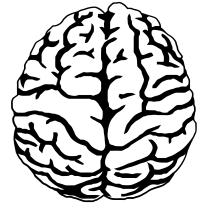
Guiding questions

- *What are the drivers of climate change?*
- *What are the impacts of climate change on ecosystems?*

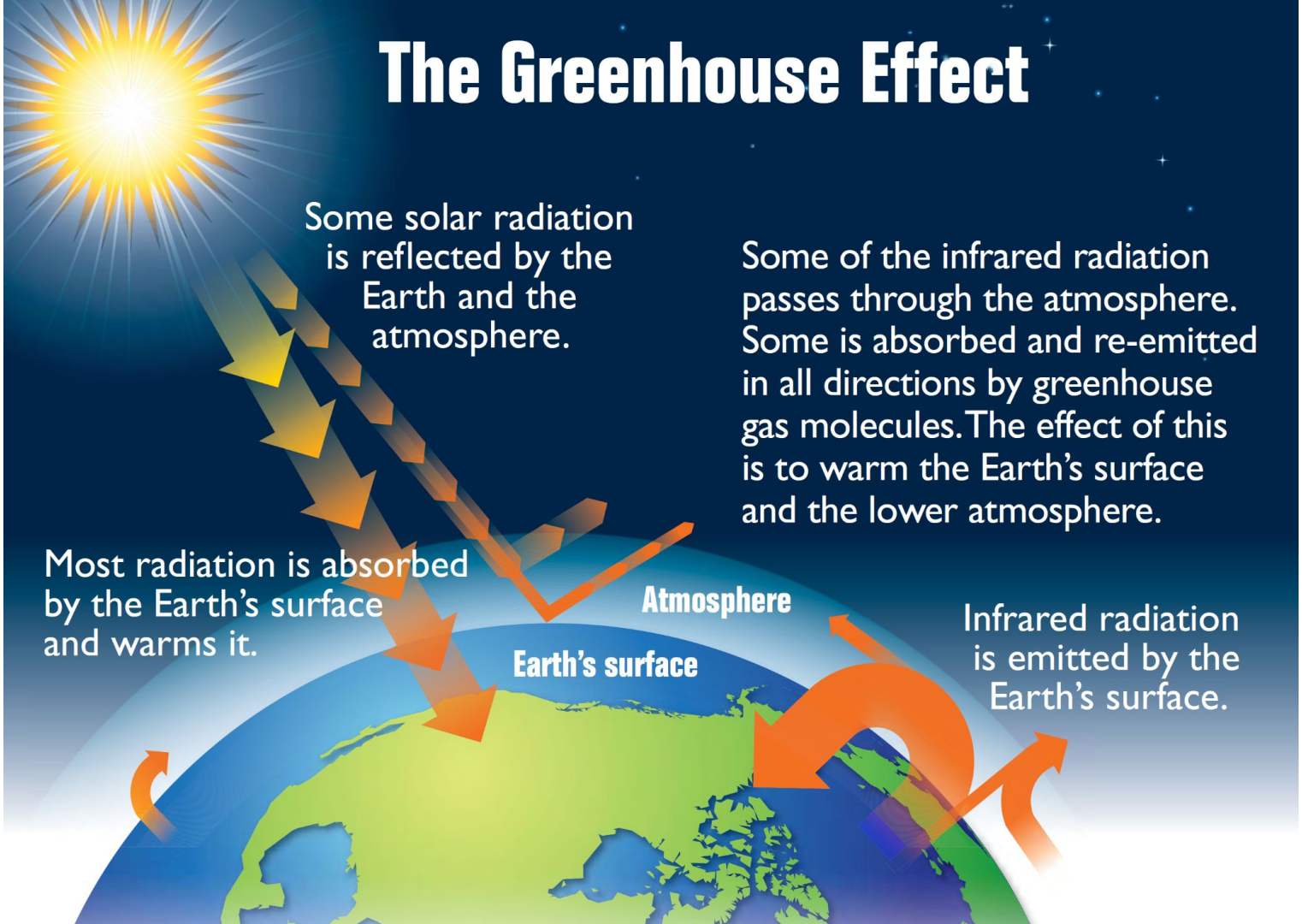
D4.3.1—Anthropogenic causes of climate change



What is the difference between global warming and climate change?



What is the “greenhouse effect”?


The Greenhouse Effect
Some solar radiation is reflected by the Earth and the atmosphere.
Some of the infrared radiation passes through the atmosphere. Some is absorbed and re-emitted in all directions by greenhouse gas molecules. The effect of this is to warm the Earth's surface and the lower atmosphere.
Most radiation is absorbed by the Earth's surface and warms it.
Atmosphere
Earth's surface
Infrared radiation is emitted by the Earth's surface.



Solar radiation is either reflected back into space or absorbed by the planet.

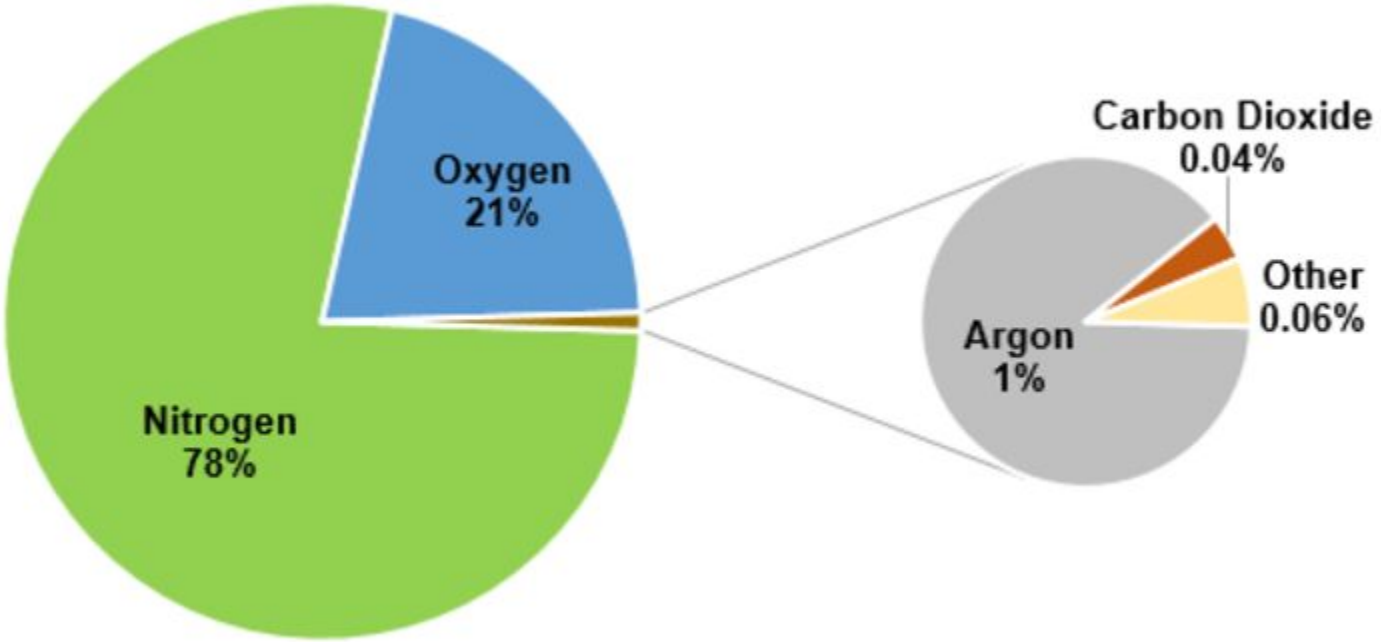
Earth reflects around 30% of sunlight (**Albedo** = 0.3).

This would lead to an average global temp of around -18℃

Actual global temperature of 15℃ due to greenhouse effect

(ice reflects sunlight (albedo 0.7), oceans absorb (albedo 0.06)... what would happen to the light reflected if the ice caps melted?)

Earth’s atmosphere



Carbon dioxide and water vapour are the most significant greenhouse gases.

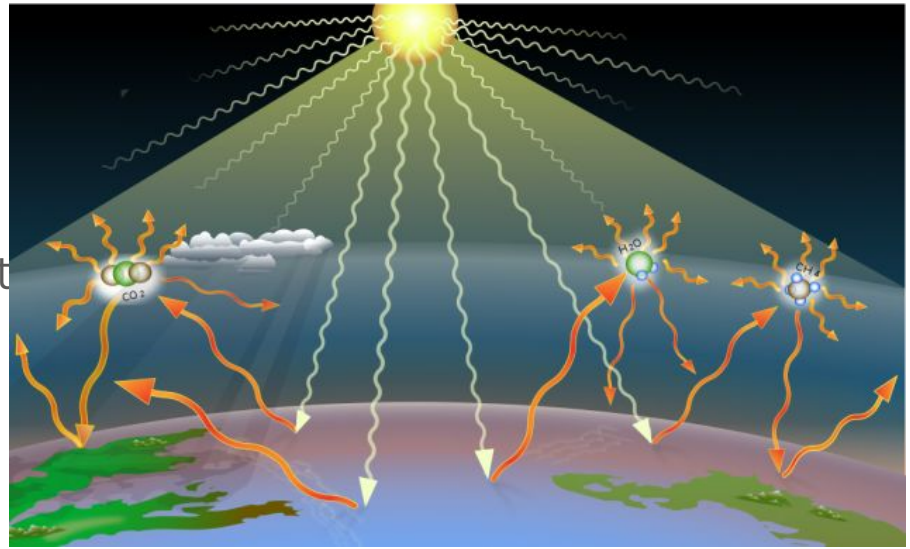
H_2O

Most abundant

Positive feedback. Hotter air, more water vapour, more greenhouse effect

CO_2

Naturally released due to respiration and volcanoes. largest anthropogenic contribution to greenhouse effect





What are the main contributors to atmospheric carbon dioxide and water vapour, and what processes reduce the concentrations?

What are the main contributors to atmospheric carbon dioxide and water vapour, and what processes reduce the concentrations?

CO2

Into atmosphere - Respiration, combustion

Out of atmosphere - Photosynthesis, dissolution in oceans

Water vapour

Into atmosphere - Evaporation, transpiration

Out of atmosphere - Precipitation

Other gases including methane and nitrogen oxides have less impact.

CH_4 (methane)

Each molecule 25x more warming than CO_2

Waterlogged sites (e.g. rice paddies), grazing cattle, landfill sites, fossil fuel extraction

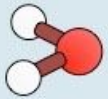
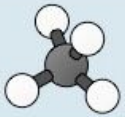
N_2O (nitrous oxide)

300x more potent than CO_2

Released by microorganisms and via agriculture and vehicle emissions

The impact of a gas depends on its ability to absorb longwave radiation as well as on its concentration in the atmosphere.

The concentrations of greenhouse gases depend upon the rate they are released and the length of time they remain in the atmosphere.

	Water	Carbon Dioxide	Methane	Nitrous Oxide
				
Atmospheric Concentration	0.01–4%*	385 ppm	1797 ppb	322 ppb
Rate of Increase	n/a	1.5 ppm/yr	7.0 ppb/yr	0.8 ppb/yr
Atmospheric Lifetime	Very short 1–5 days	Variable 5–200 yr	12 yr	120 yr
Global Warming Potential (GWP)	n/a†	1	21	310

* The amount of water vapor in the air varies according to temperature and density of air (usually ~1–3% of troposphere)

† Water vapor levels vary strongly according to region, so rates of change and warming potential cannot be assessed

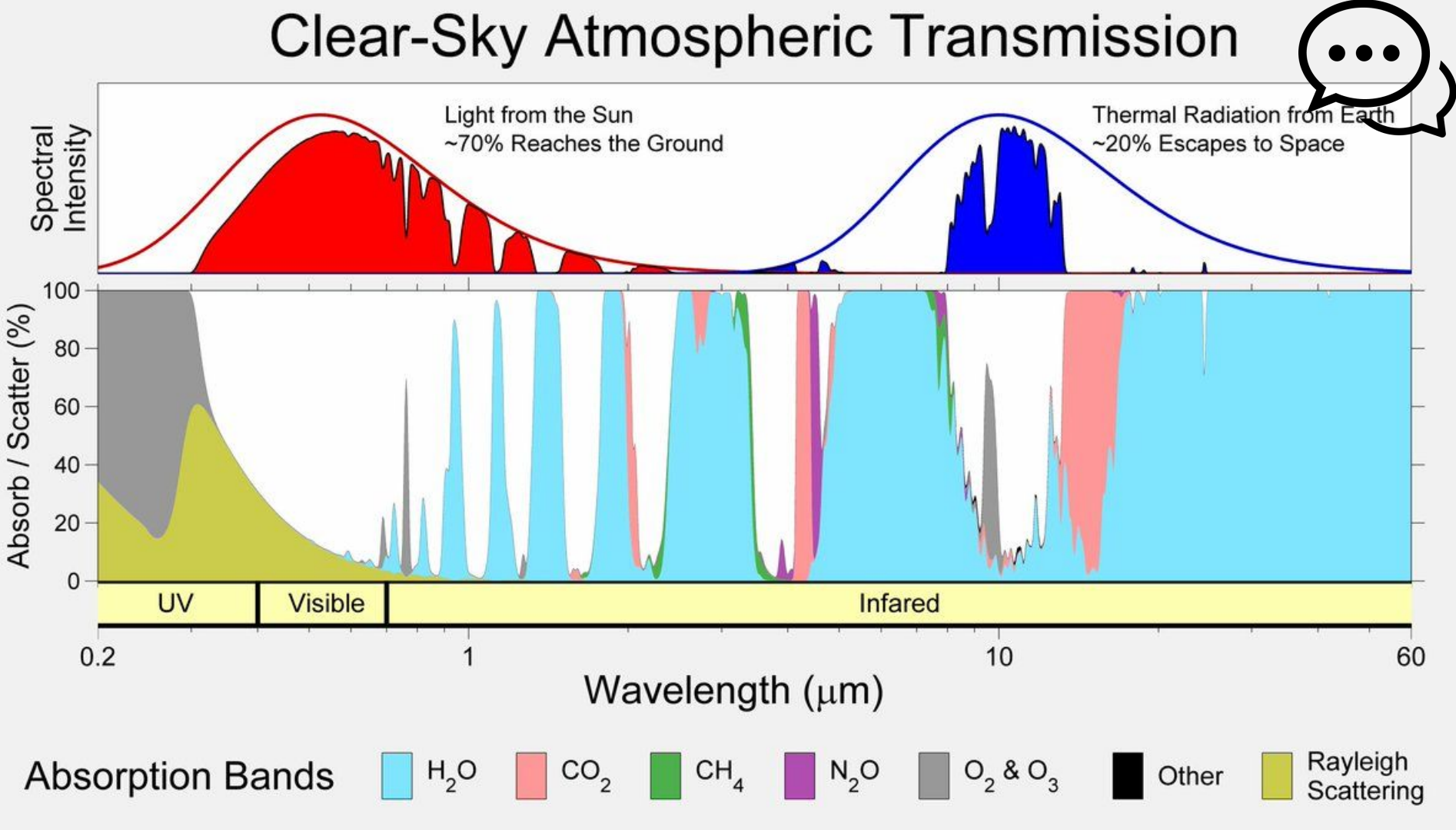
The warmed Earth emits longer wavelength radiation (heat).

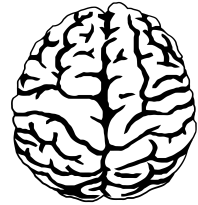
The faster a molecule vibrates the shorter the wavelength of the energy it emits.

As the earth is cooler than the sun it radiates energy at a longer wavelength.

Describe and interpret the following graph

Clear-Sky Atmospheric Transmission





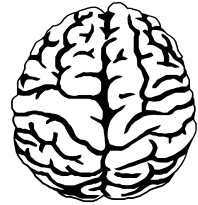
What is the difference between the weather and the climate?

What is the difference between the weather and the climate?

Weather is short term changes in the elements (temperature, rain, wind) (hour to hour, day to day)

Climate is long term changes in the weather (year to year, decade to decade)

Climate systems



How do we measure historical climate?

Measuring past climate

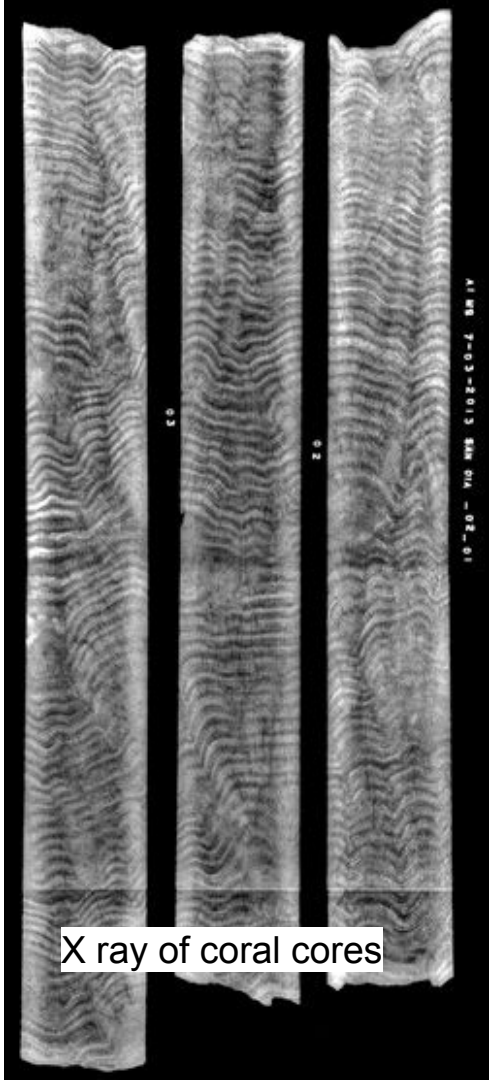
Recent and historical records

Tree rings

Corals

Ice cores





X ray of coral cores

Ice cores

Up to 850,000 years of climate history



Ratio of oxygen isotopes ^{18}O:^{16}O

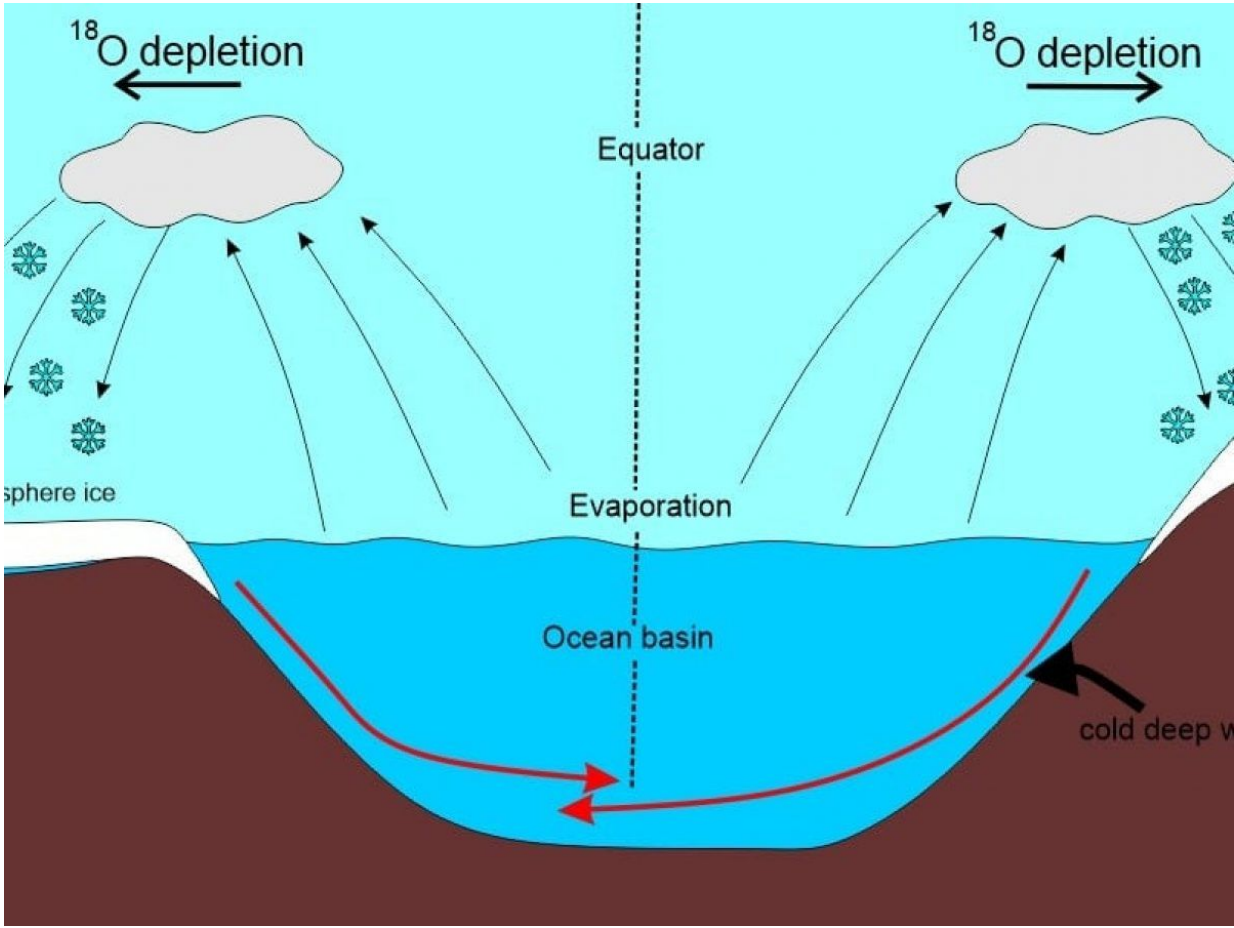
^{18}O more difficult to evaporate than ^{16}O.

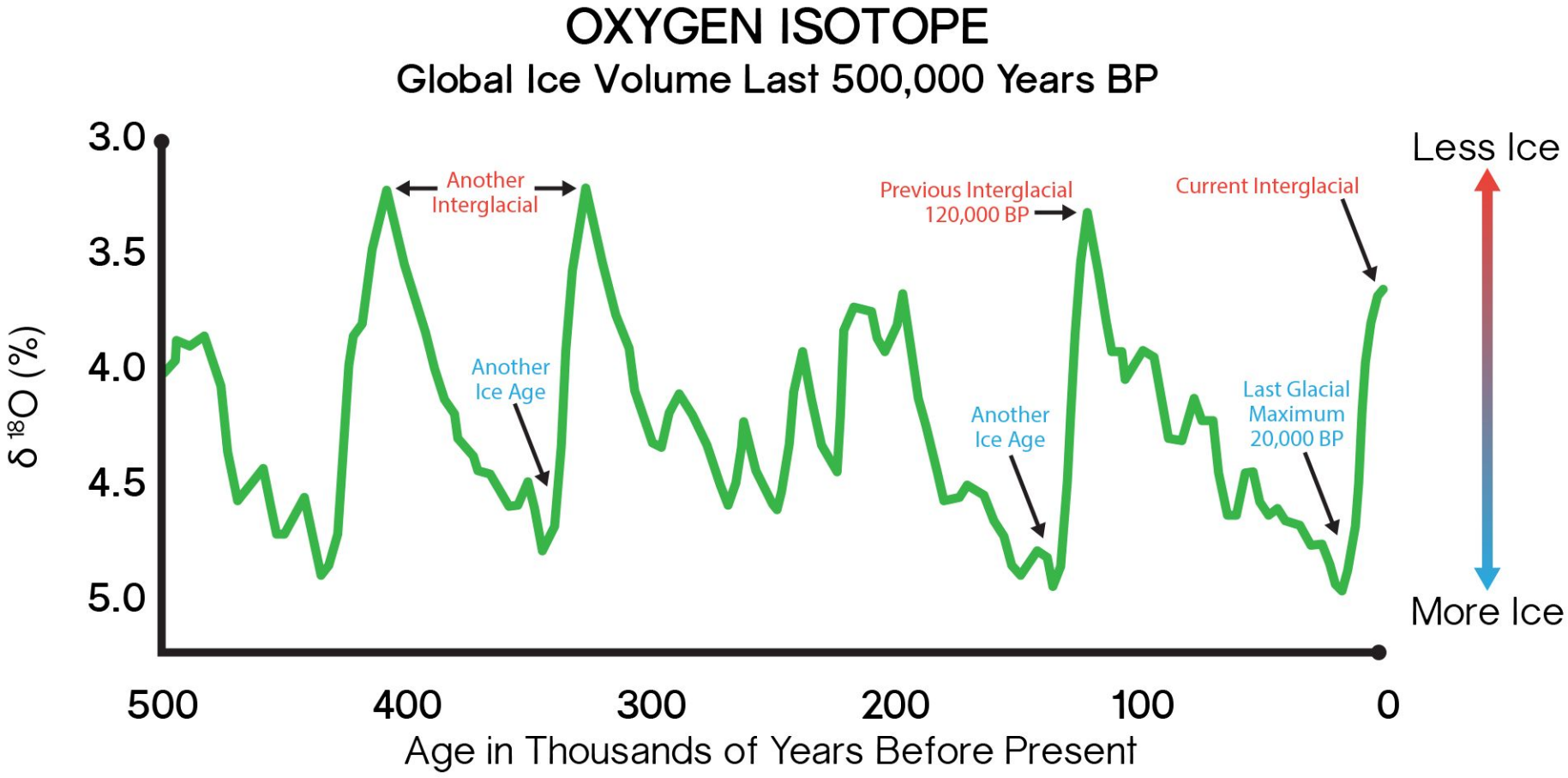
Warm temperature encourages relatively more ^{18}O

Ratio therefore can show temperature.

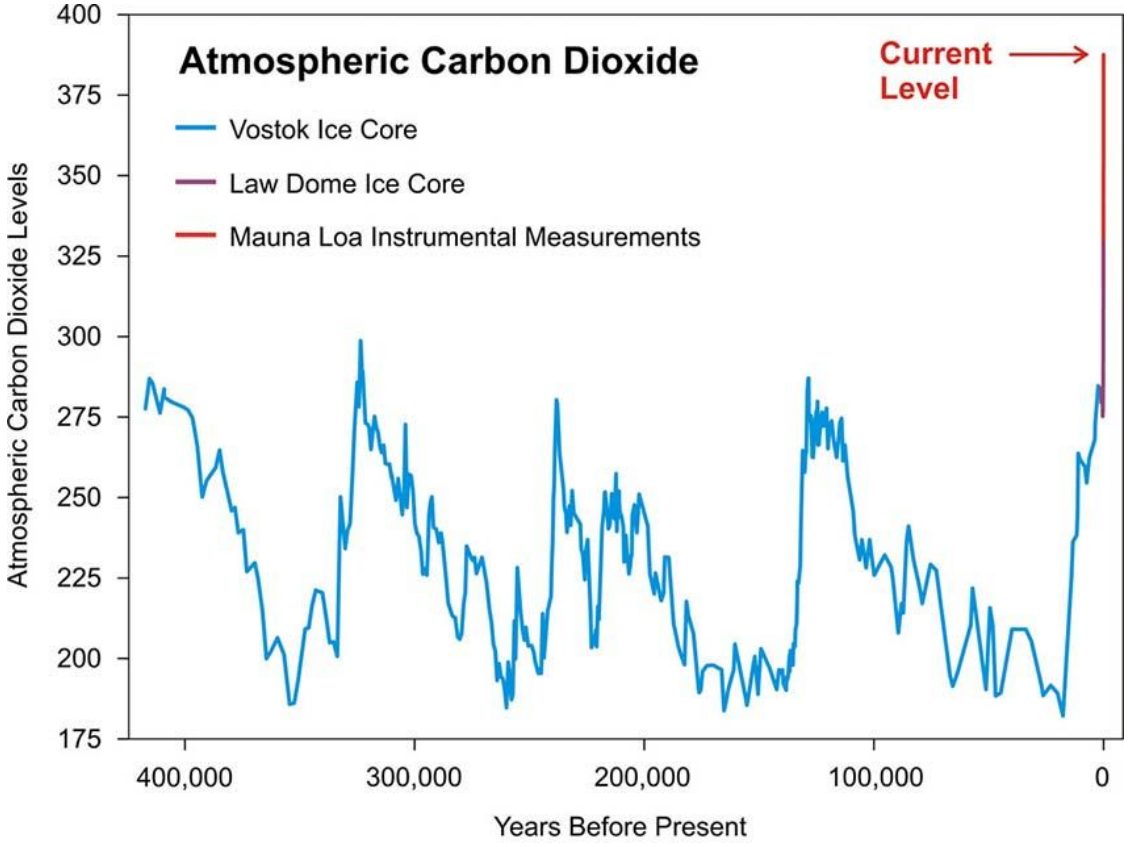
By the time water vapour falls as snow at the poles it is depleted of ^{18}O (which is harder to vaporise and more likely to precipitate)

Warmer climates lead to a higher concentration of ^{18}O in atmosphere and therefore in snow.

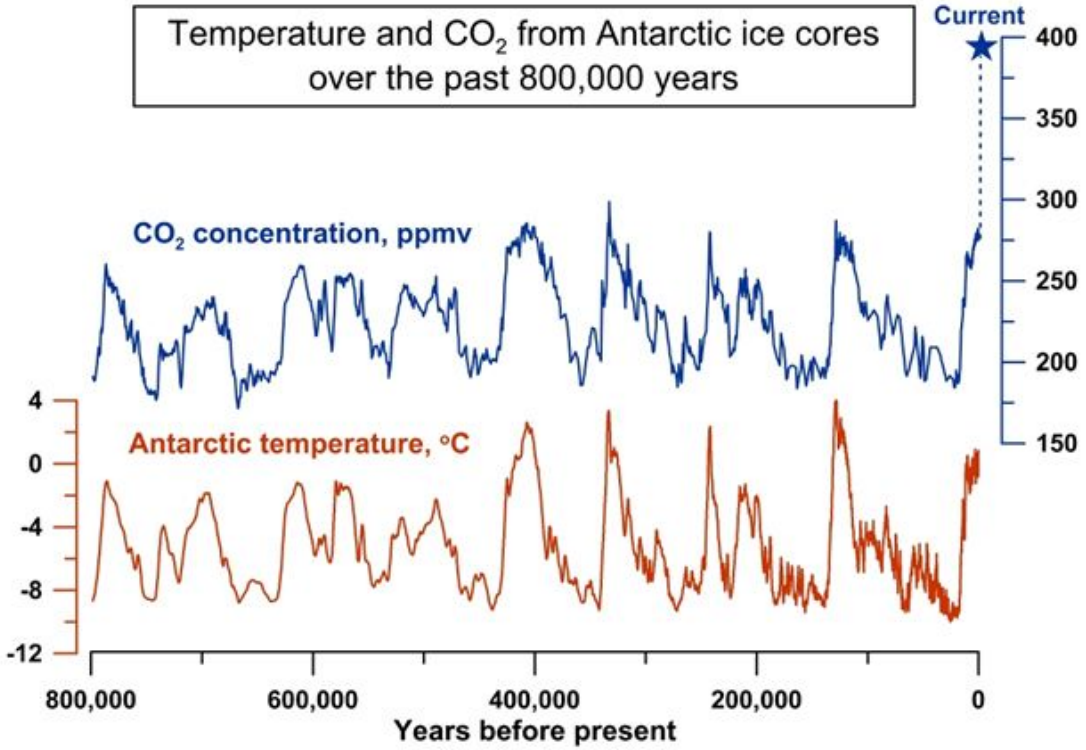



OXYGEN ISOTOPE
Global Ice Volume Last 500,000 Years BP
3.0
3.5
4.0
4.5
5.0
δ 18O (%)
Another Interglacial
Another Ice Age
Previous Interglacial 120,000 BP
Another Ice Age
Current Interglacial
Last Glacial Maximum 20,000 BP
Less Ice
More Ice
500
400
300
200
100
0
Age in Thousands of Years Before Present

Ice cores also contain trapped air bubbles that represent historical atmospheric samples.



There is clearly a correlations between global temperatures and carbon dioxide concentrations on Earth.

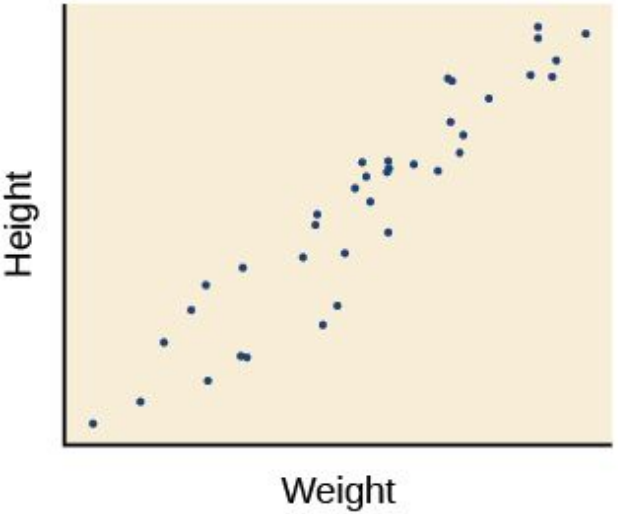


Correlation or causation?

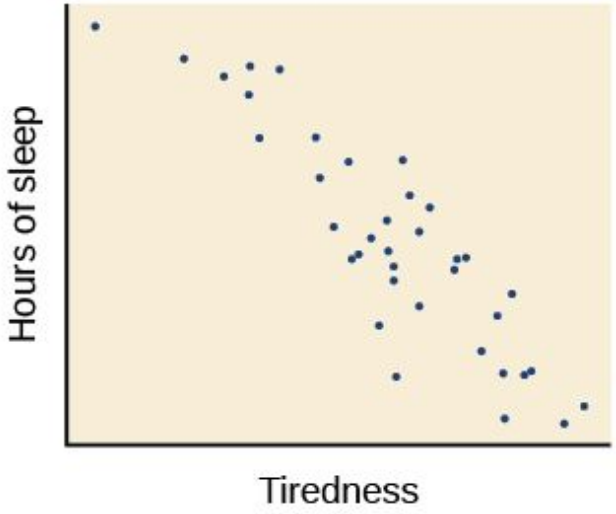


From these graphs can you confidently assert that rising CO2 is the cause of climate change?

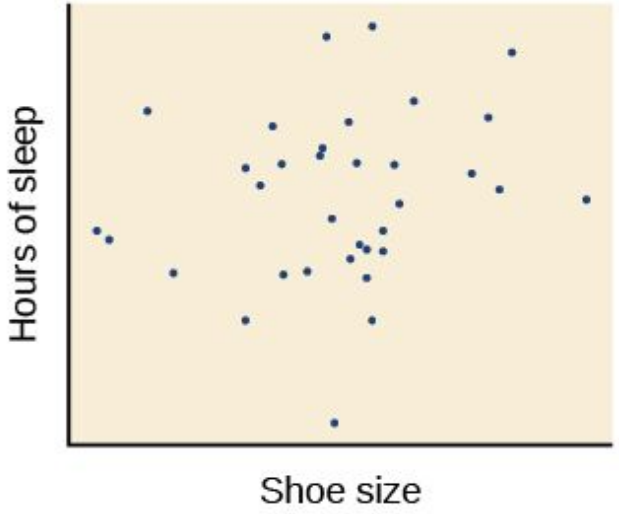
Types of correlation



(a) Positive Correlation



(b) Negative Correlation

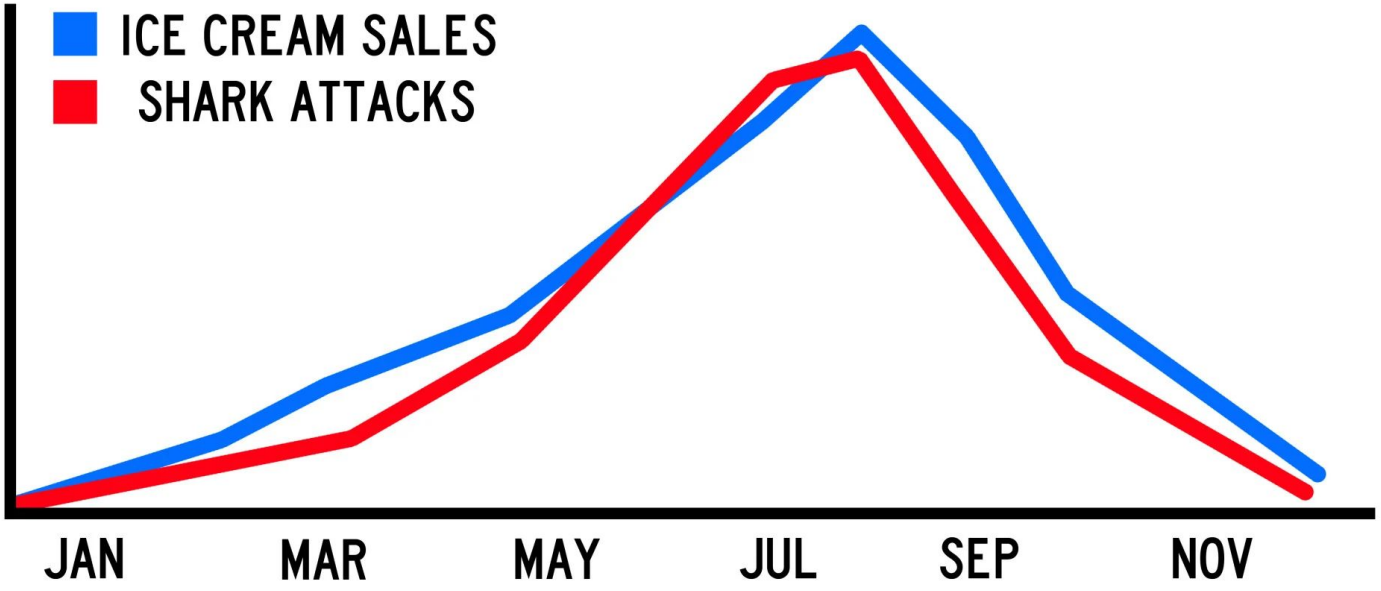


(c) No Correlation

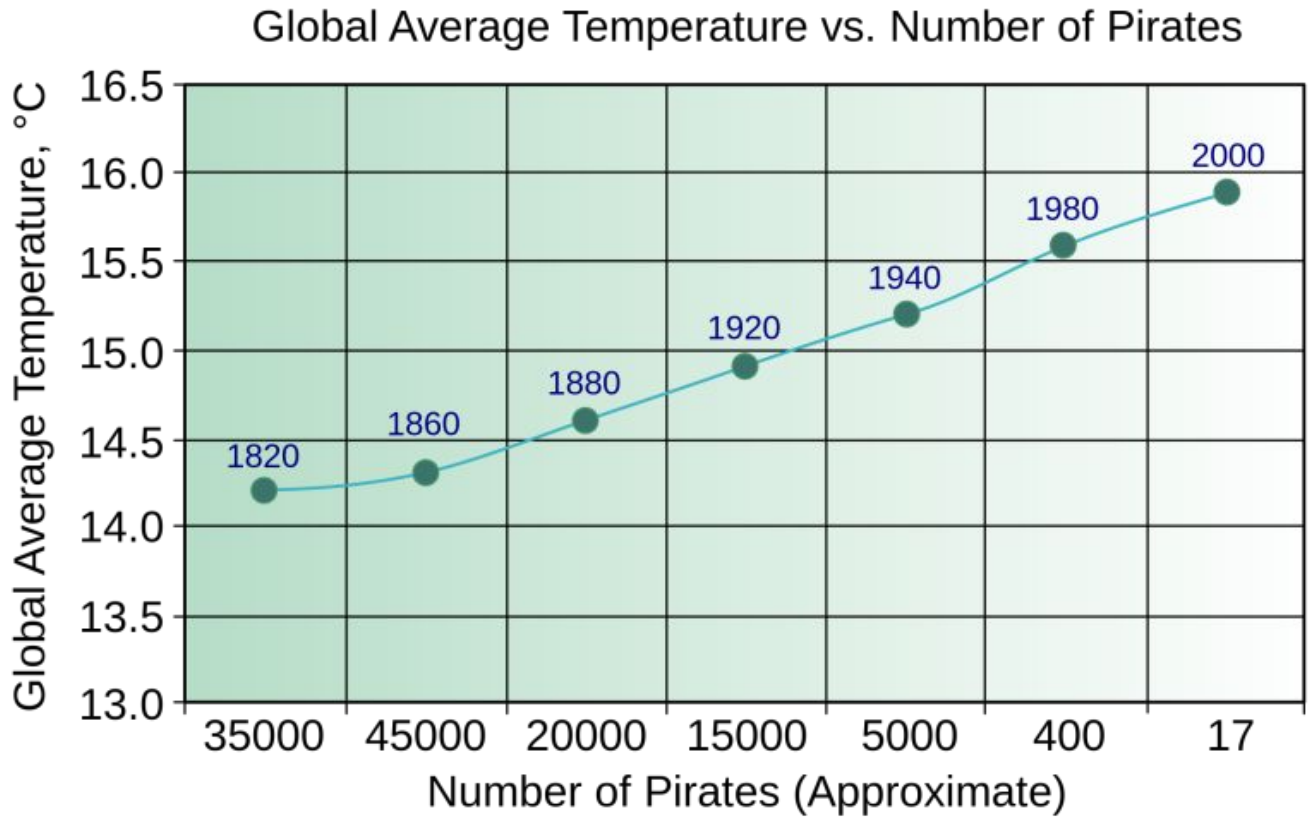
Do ice cream sales cause shark attacks?



CORRELATION IS NOT CAUSATION!



Do pirates cause global warming?




SOMEDAY, I WANT TO GET MARRIED BECAUSE STUDIES SHOW THAT MARRIED PEOPLE ARE HAPPIER.
Dilbert.com DilbertCartoonist@gmail.com
A SMARTER INTERPRETATION IS THAT NO ONE WANTS TO MARRY AN UNHAPPY PERSON.
2-10-12 ©2012 Scott Adams, Inc. /Dist. by Universal Uclick



When could you infer causation from a pattern of correlation?

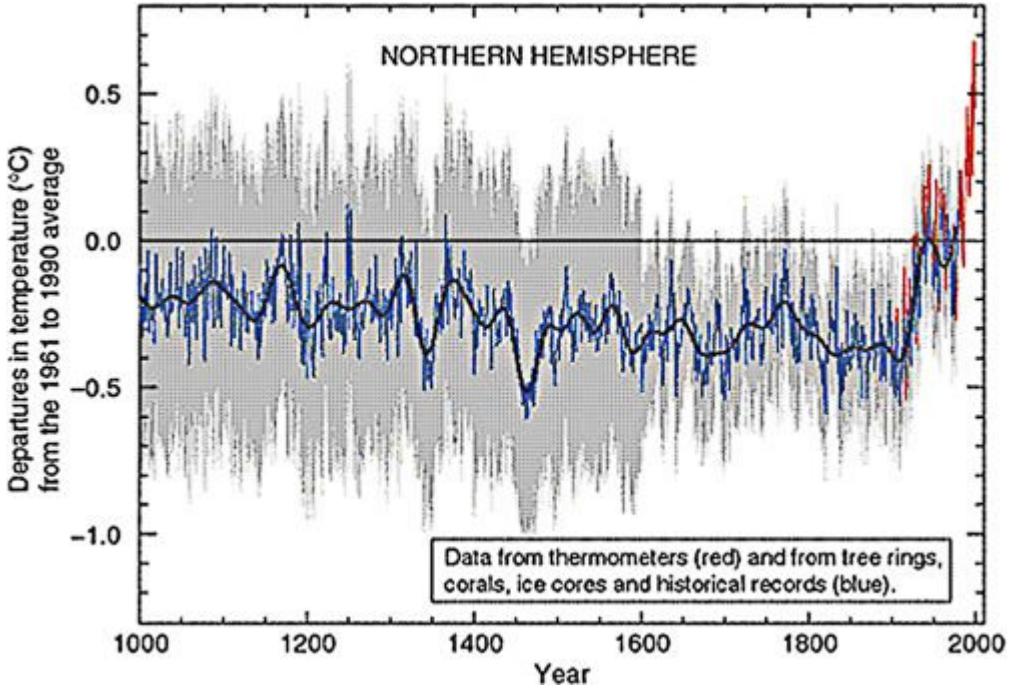
When could you infer causation from a pattern of correlation?

When there is a clear evidence based explanation!

Nature of Science

Students should be able to distinguish between positive and negative correlation and should also distinguish between correlation and causation. For example, data from Antarctic ice cores shows a positive correlation between global temperatures and atmospheric carbon dioxide concentrations over hundreds of thousands of years. This correlation does not prove that carbon dioxide in the atmosphere increases global temperatures, although other evidence confirms the causal link.

Hockey stick graph



First published in 1998



What was happening around 1900 to account for this change in temperature?

Global temperatures and climate patterns are influenced by concentrations of greenhouse gases.

There is a lot of evidence to suggest that increased concentrations of greenhouse gasses causes the planet to warm.

Leads to:

Destabilised climate patterns

Extreme weather events

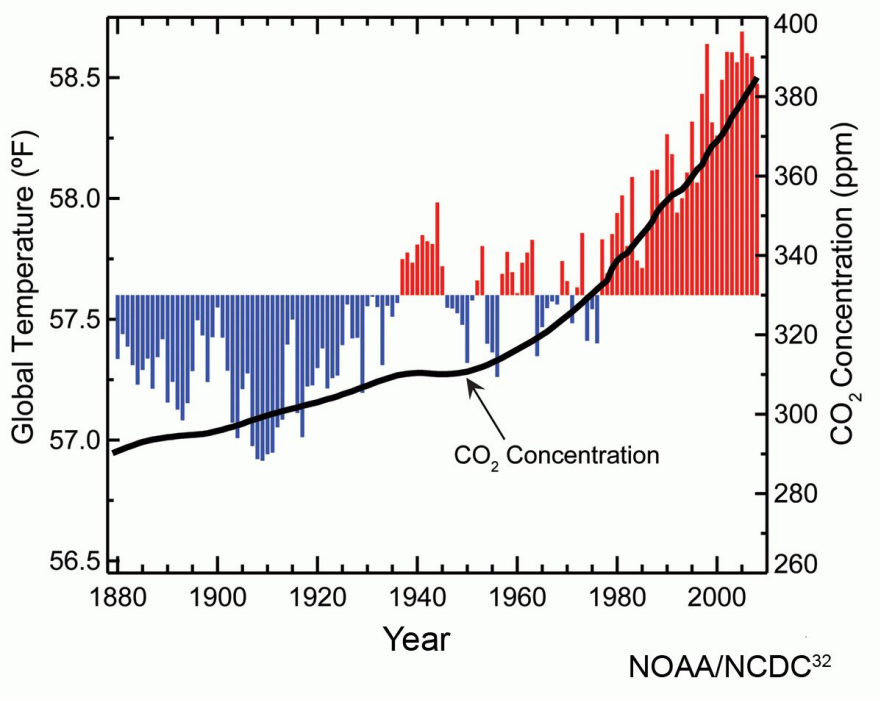
Changes in air and ocean current systems

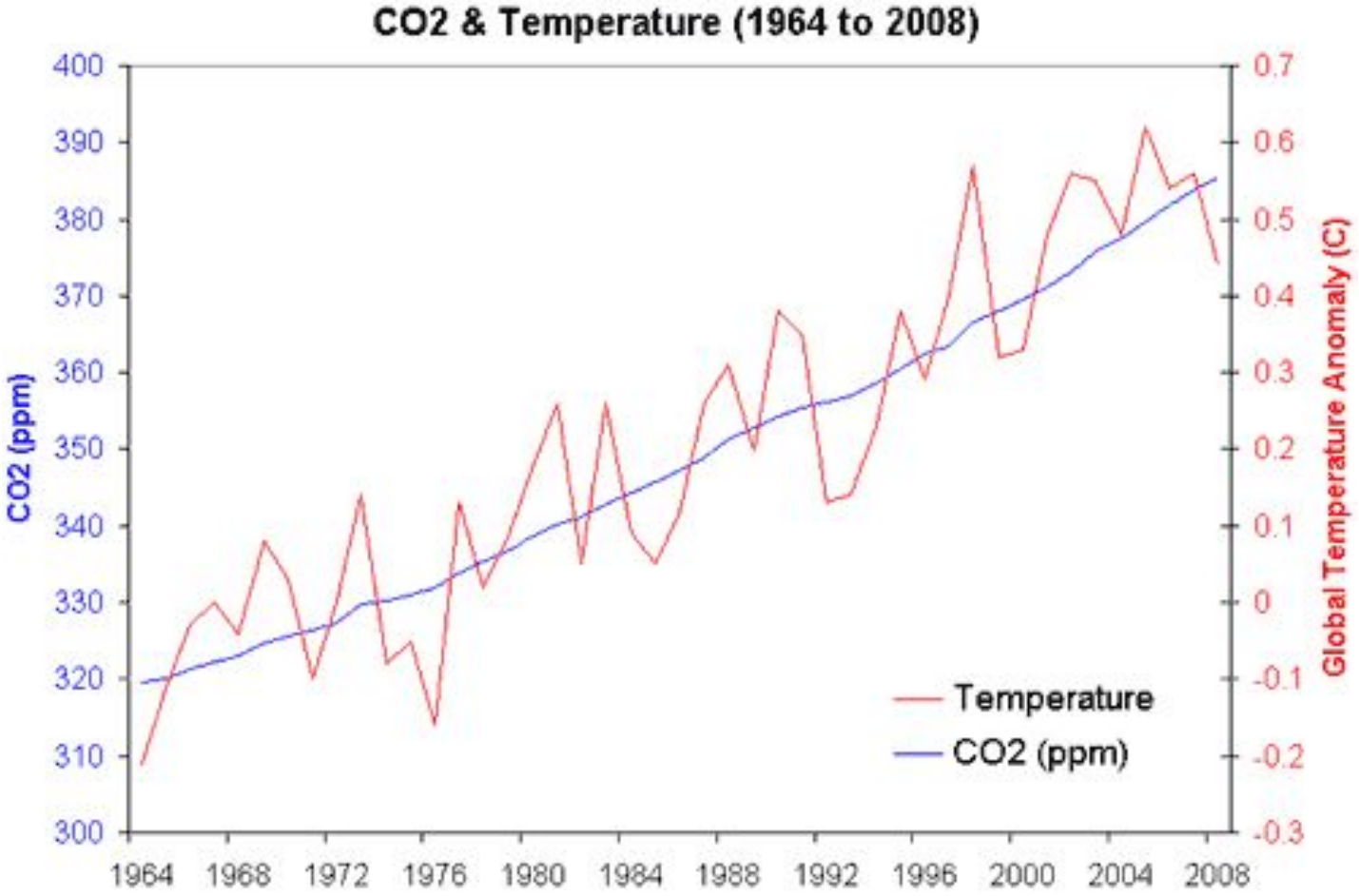
Floods and droughts

There is a correlation between rising atmospheric concentrations of carbon dioxide since the start of the industrial revolution 200 years ago and average global temperatures.

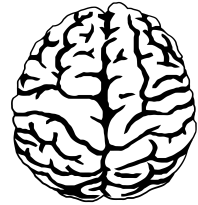
Historic atmospheric concentrations of CO2 fluctuated dramatically but never exceeded 300ppm.

Currently at around 420ppm




CO2 & Temperature (1964 to 2008)
CO2 (ppm)
Global Temperature Anomaly (C)
400
390
380
370
360
350
340
330
320
310
300
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
-0.1
-0.2
-0.3
1964
1968
1972
1976
1980
1984
1988
1992
1996
2000
2004
2008
Temperature
CO2 (ppm)

What are the anthropological causes of global warming?



What are the anthropological causes of global warming?

Burning fossil fuels

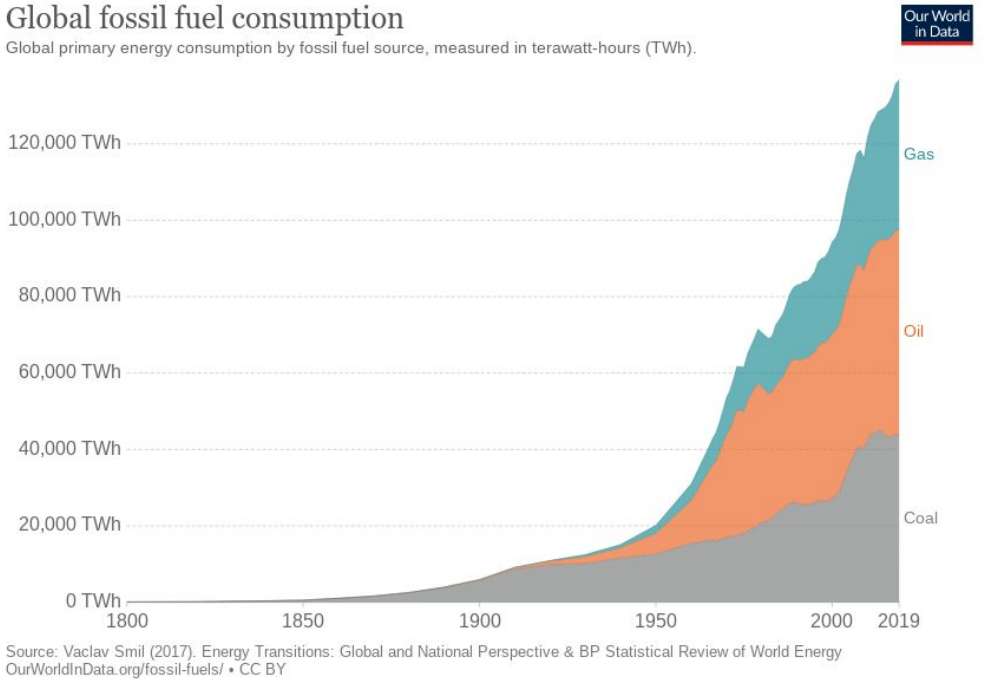
Deforestation and tree clearing

Agriculture

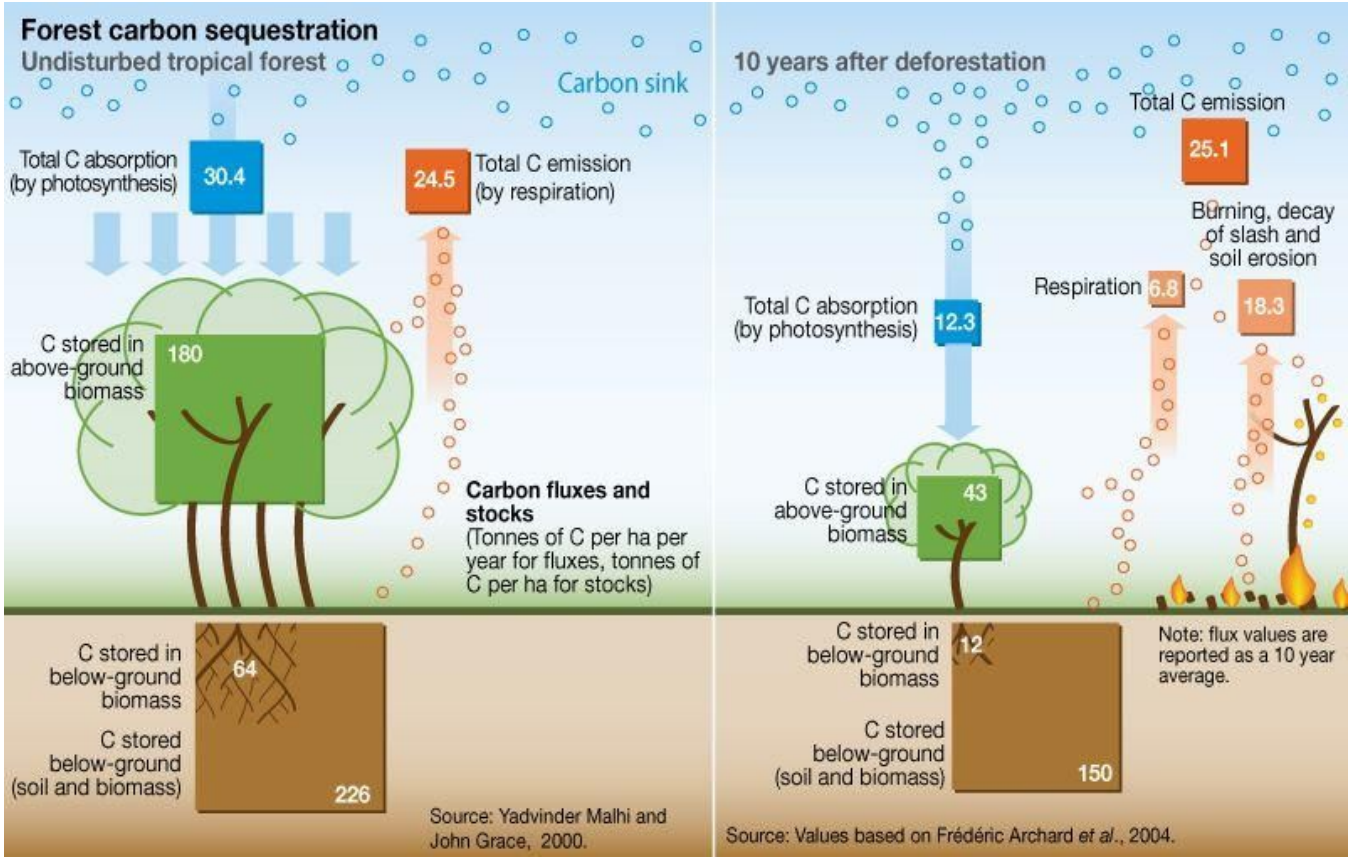
Fossil fuels

Recent increases in atmospheric carbon dioxide are largely due to increases in the combustion of fossil fuels.

The industrial revolution lead the way to the energy hungry lifestyles of the 21st century.



Deforestation

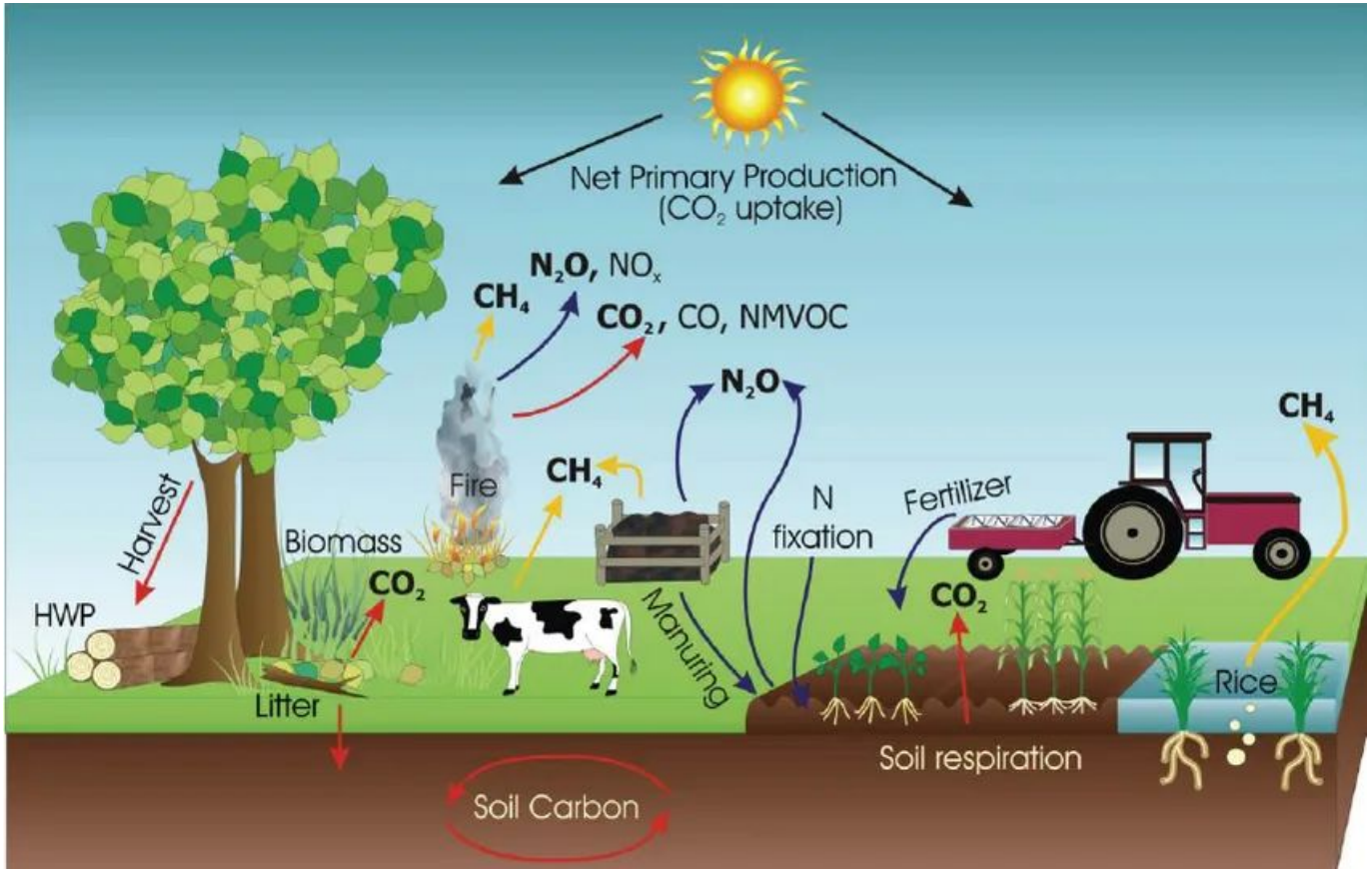


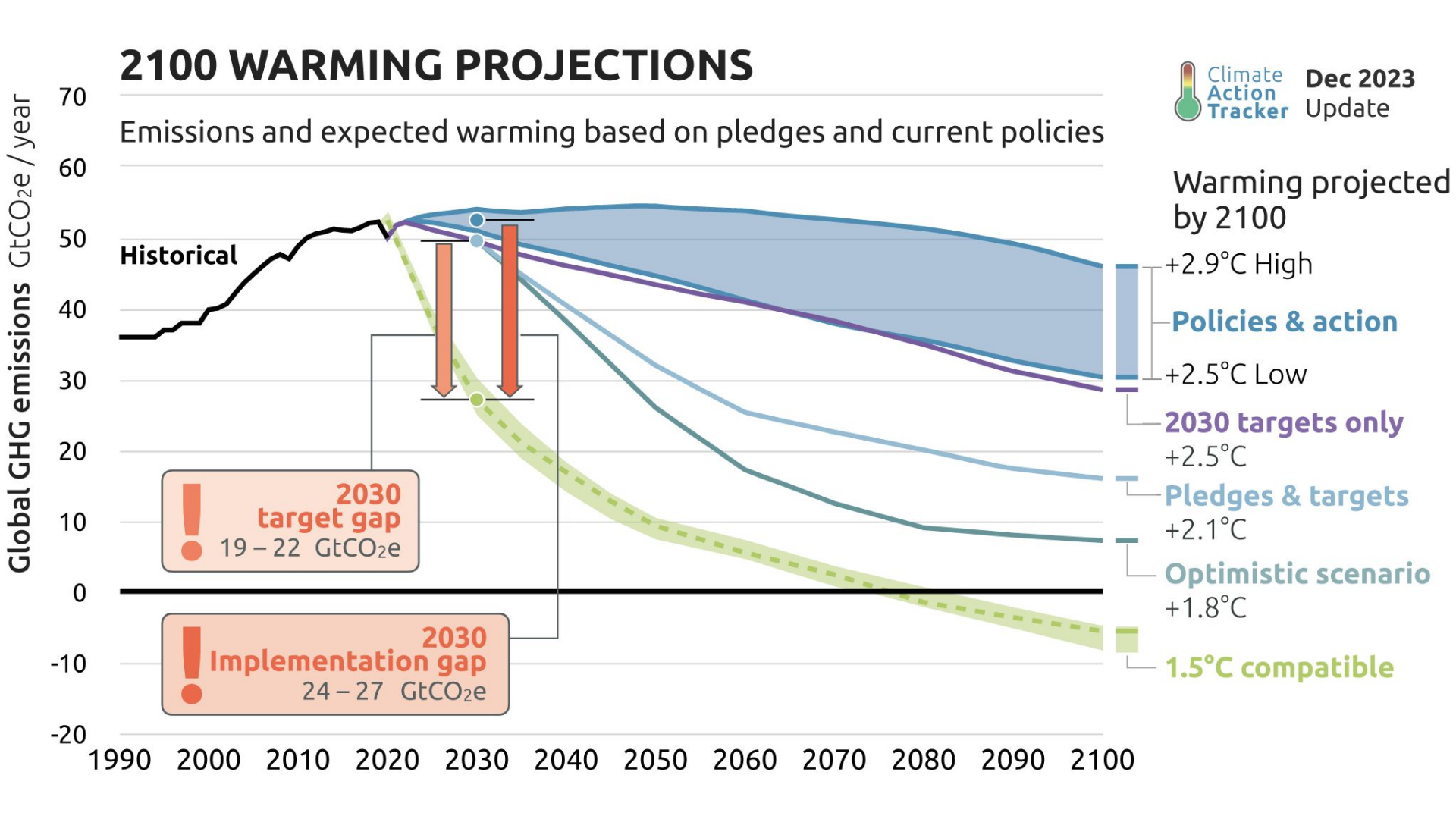
Agriculture

From this image,list as many factors as you can relating to agriculture and climate change.



Agriculture




2100 WARMING PROJECTIONS
Emissions and expected warming based on pledges and current policies
Climate Action Tracker
Dec 2023 Update
Global GHG emissions GtCO2e / year
70
60
50
40
30
20
10
0
-10
-20
1990
2000
2010
2020
2030
2040
2050
2060
2070
2080
2090
2100
Historical
2030 target gap
19 – 22 GtCO2e
2030 Implementation gap
24 – 27 GtCO2e
Warming projected by 2100
+2.9°C High
Policies & action
+2.5°C Low
2030 targets only
+2.5°C
Pledges & targets
+2.1°C
Optimistic scenario
+1.8°C
1.5°C compatible


The
Economist
What 3°C of
global warming looks like

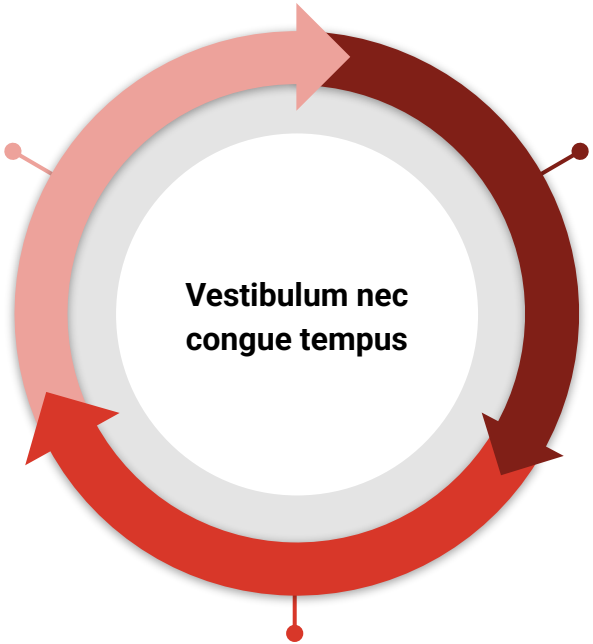
D4.3.2—Positive feedback cycles in global warming


NOW
THIS
The Climate
Tipping Points

Climate Feedback Loops can be Positive or Negative



Release of more greenhouse gases

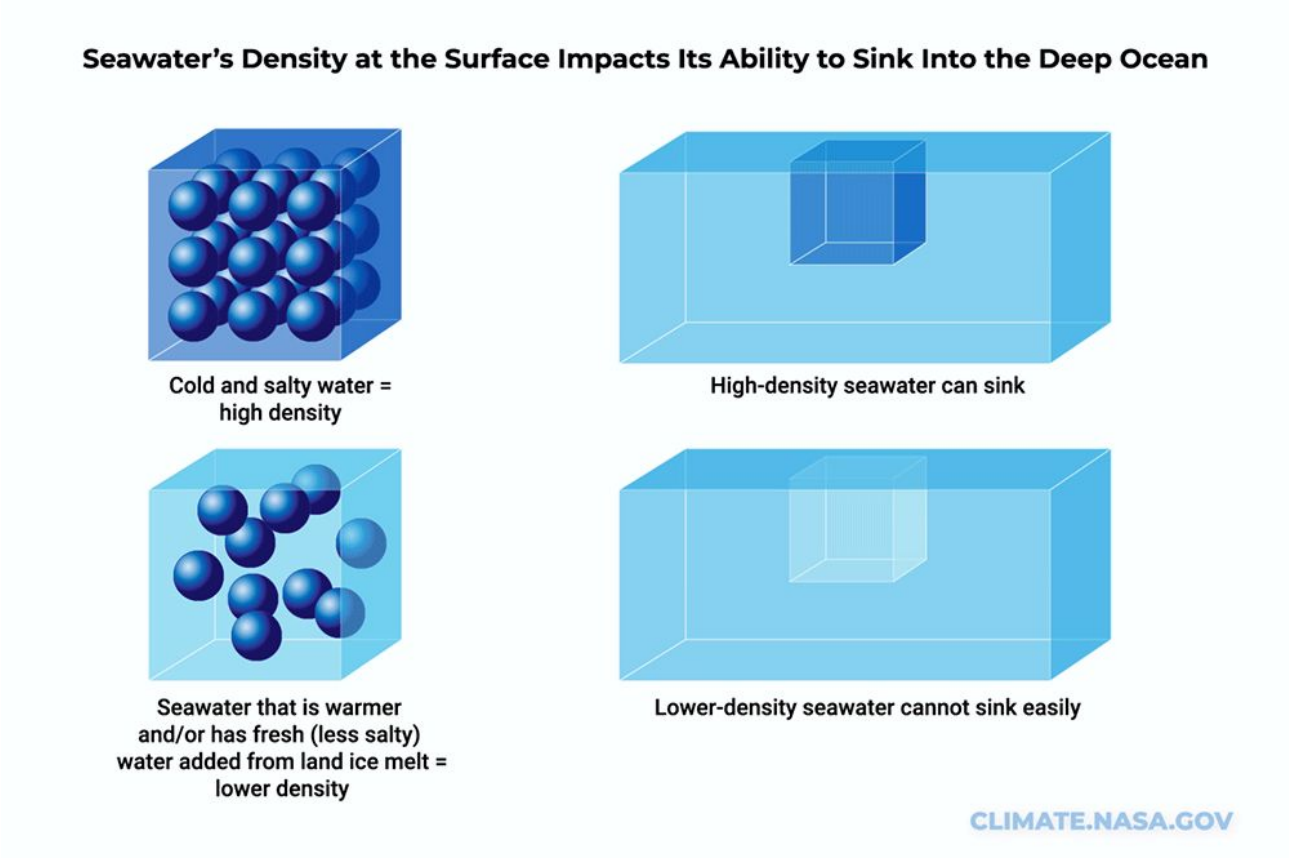


Increase in atmospheric greenhouse gases

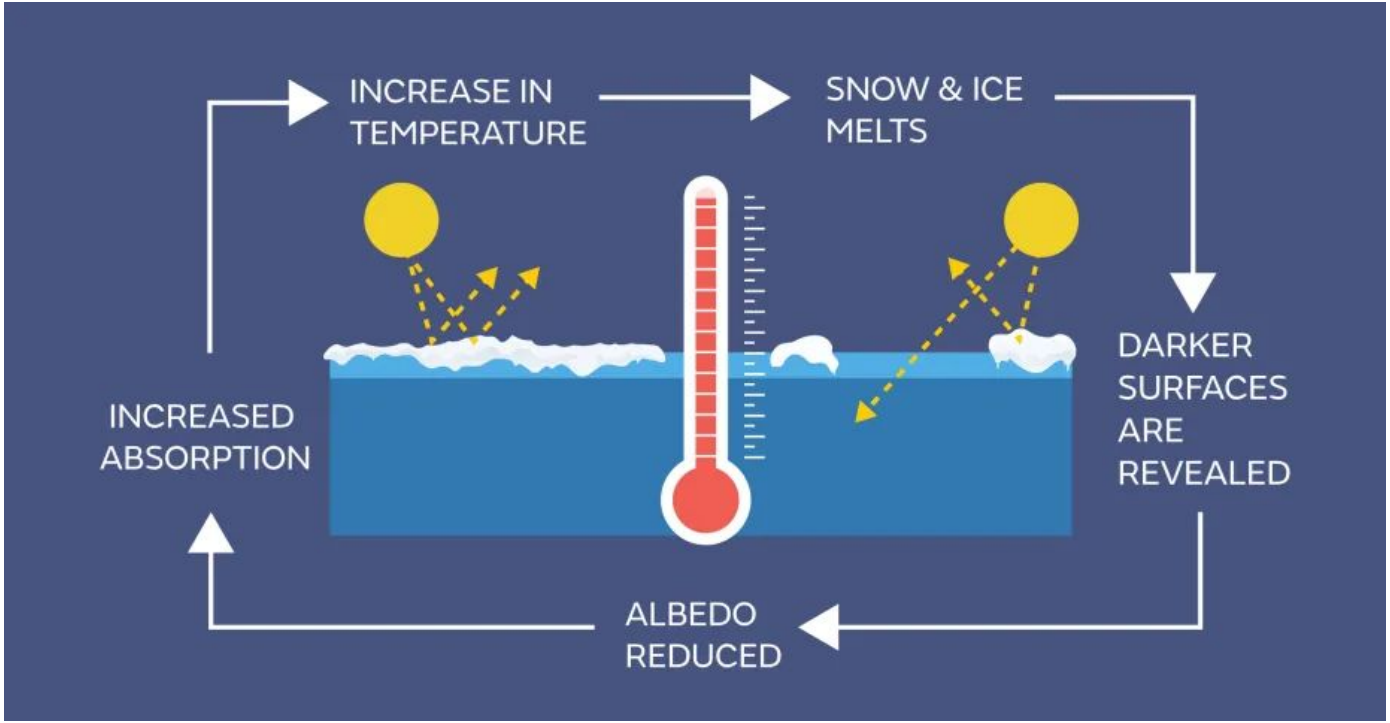
Increase in global temperature

Release of carbon dioxide from deep ocean

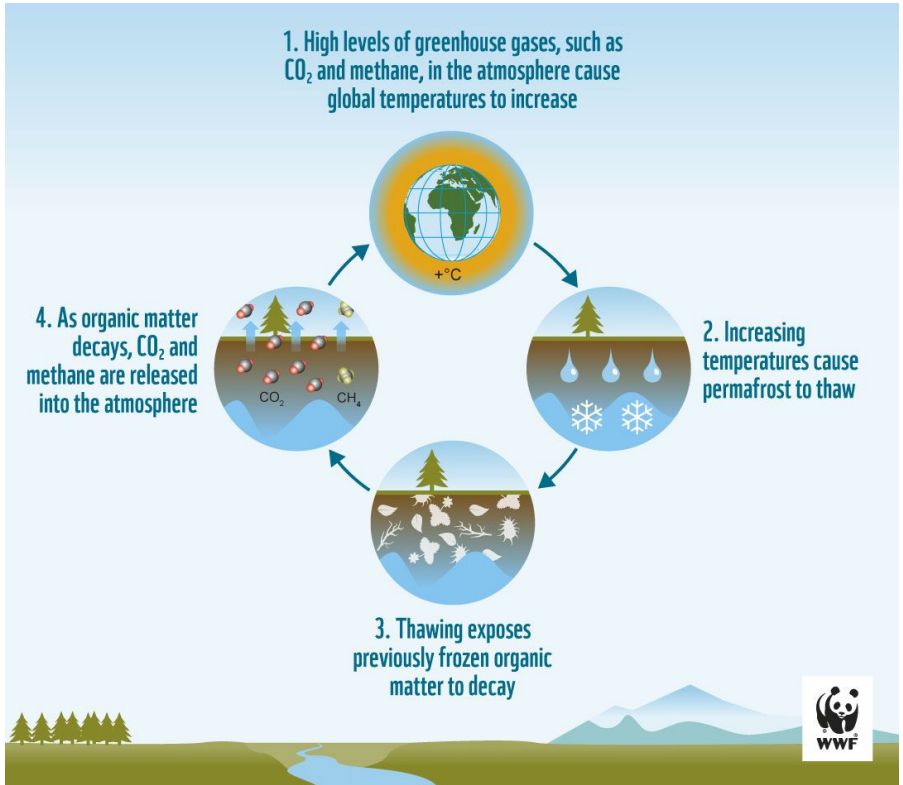
Less deep water storage of CO_2, more surface CO_2 and therefore less absorbed from atmosphere!



Increases in absorption of solar radiation due to loss of reflective snow and ice



Accelerating rates of decomposition of peat and previously undecomposed organic matter in permafrost

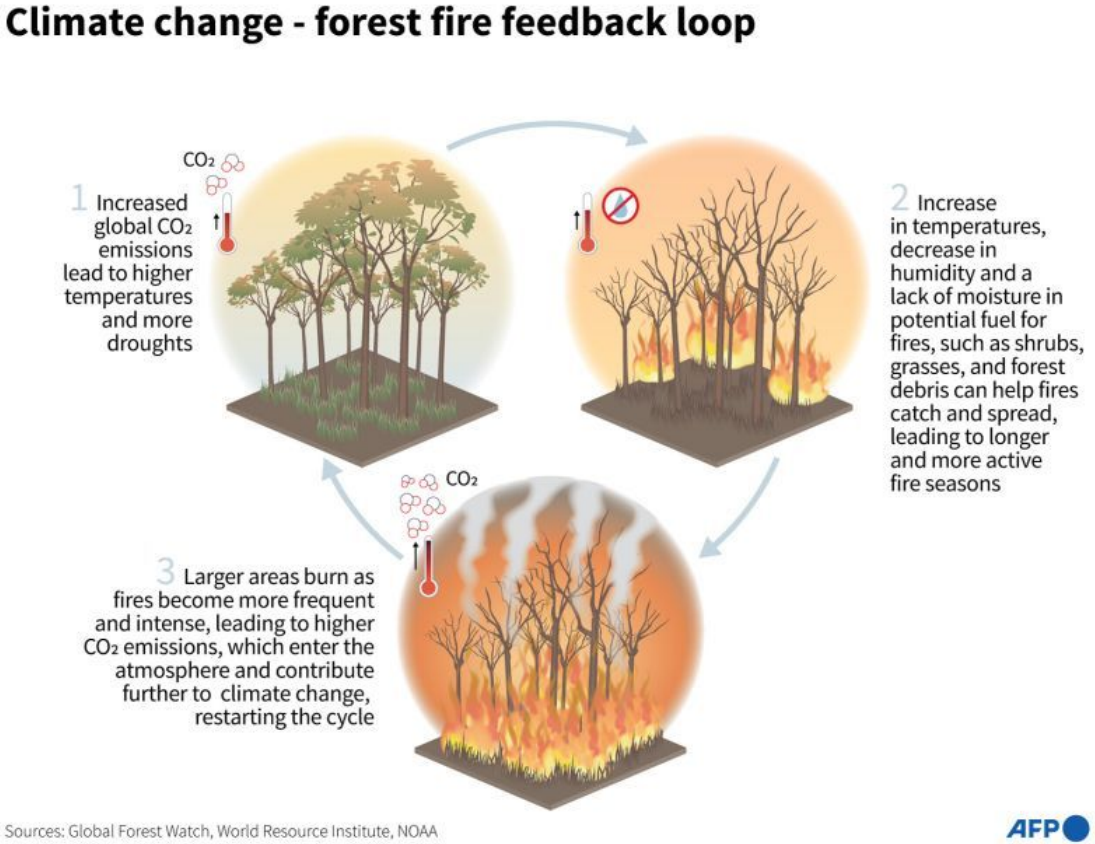


Release of methane from melting permafrost



Increase in droughts and forest fires

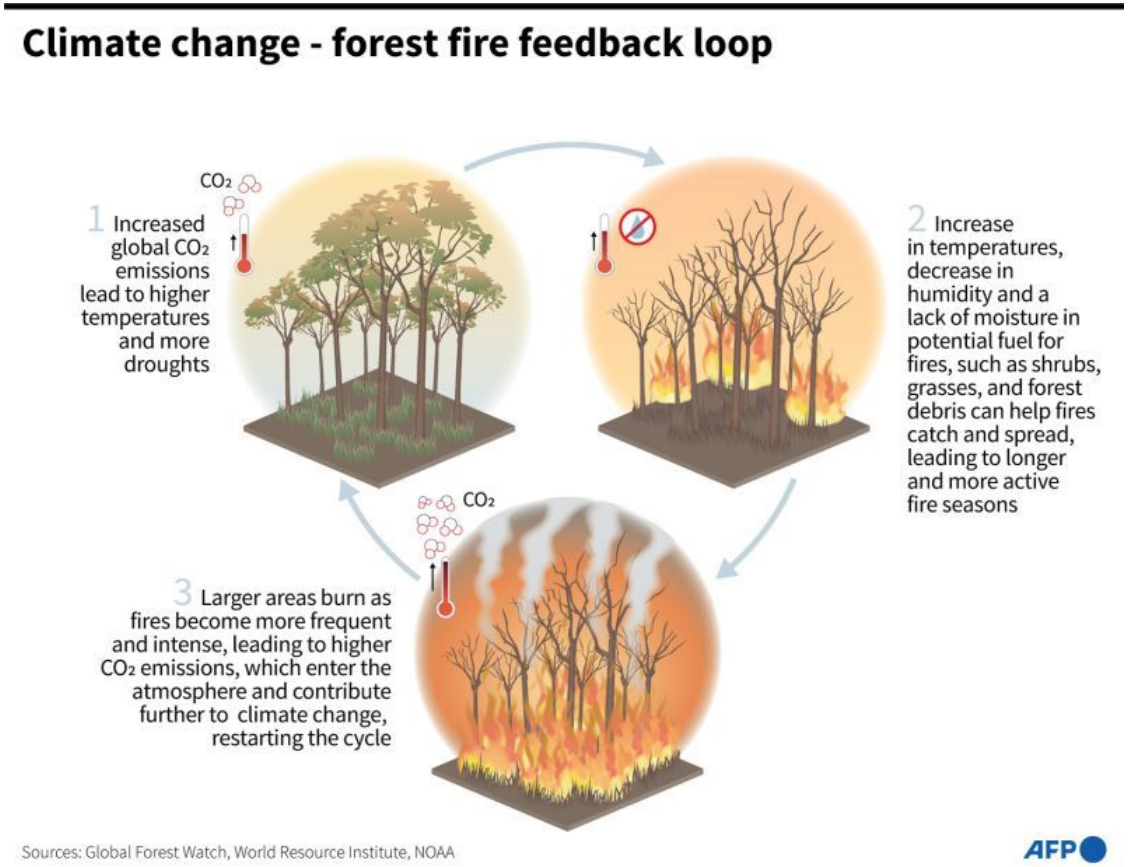
How else do forest fires and drought lead to increase atmospheric CO_2?



Increase in droughts and forest fires

How else do forest fires and drought lead to increase atmospheric CO_2?

Fewer plants = less photosynthesis!



D4.3.3—Change from net carbon accumulation to net loss in boreal forests as an example of a tipping point

Warmer temperatures

Decreased snowfall

Increased drought

Increased wildfires

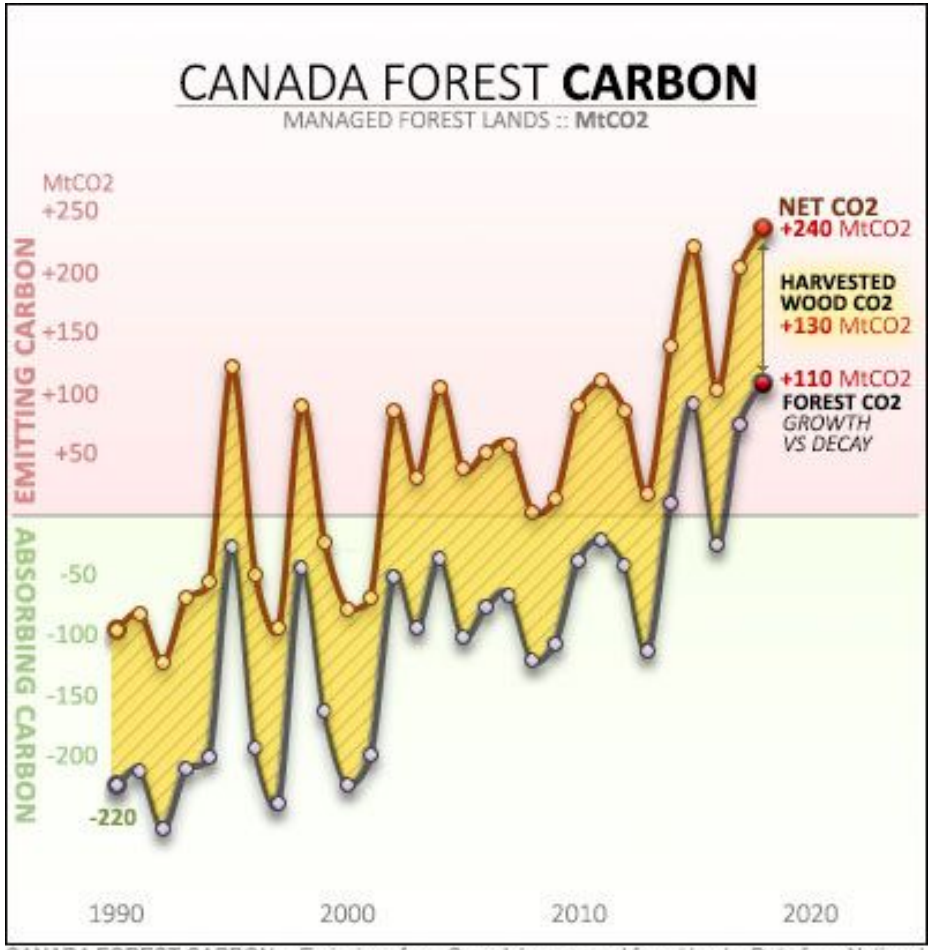
Decreased tree cover

Increased CO_2 release

Decreased carbon storage



Discuss this graph



CANADA FOREST CARBON -- Emissions from Canada's managed forest lands. Data from National Inventory Reports 2019 & 2020, with full time series for some tables provided to author by Environment Canada. Black line is net forest flux (table 6-5). Yellow area is HWP (table 10s1). Brown top line is sum of two. CHART by Barry Saxifrage at VisualCarbon.org & NationalObserver.com. May 2020

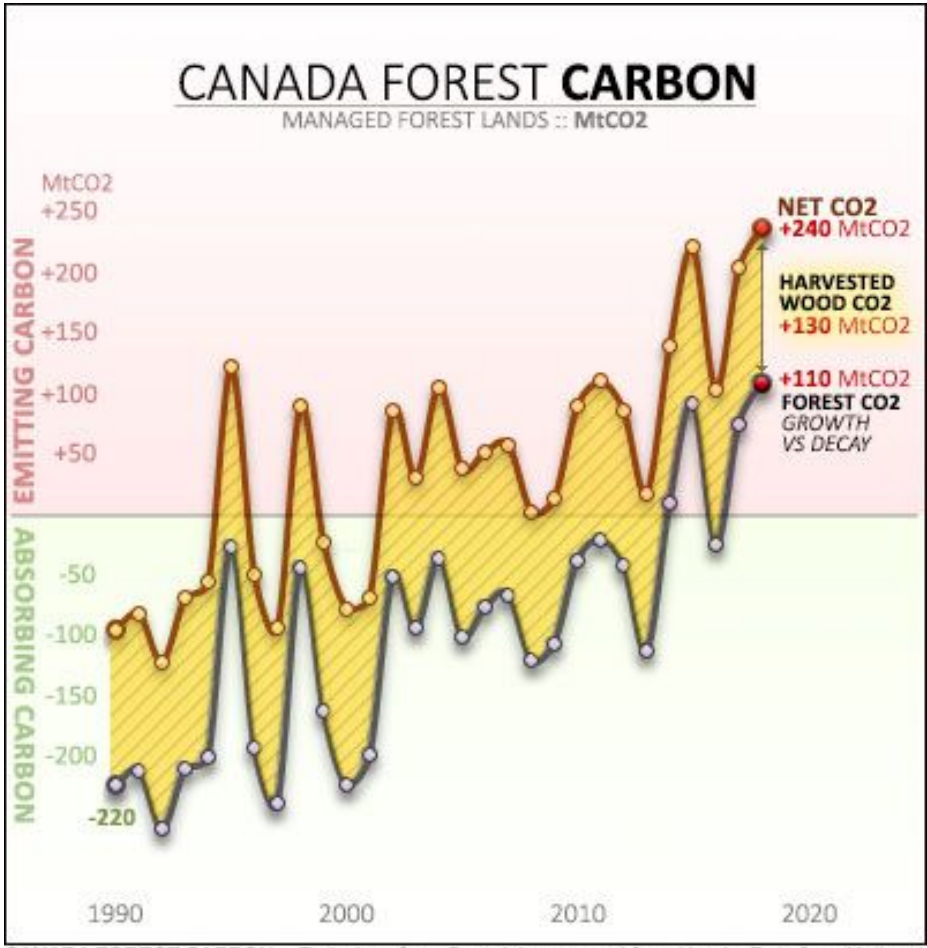
Discuss this graph

Growth vs decay

- Wildfires
- Insect kills
- Slash and burn

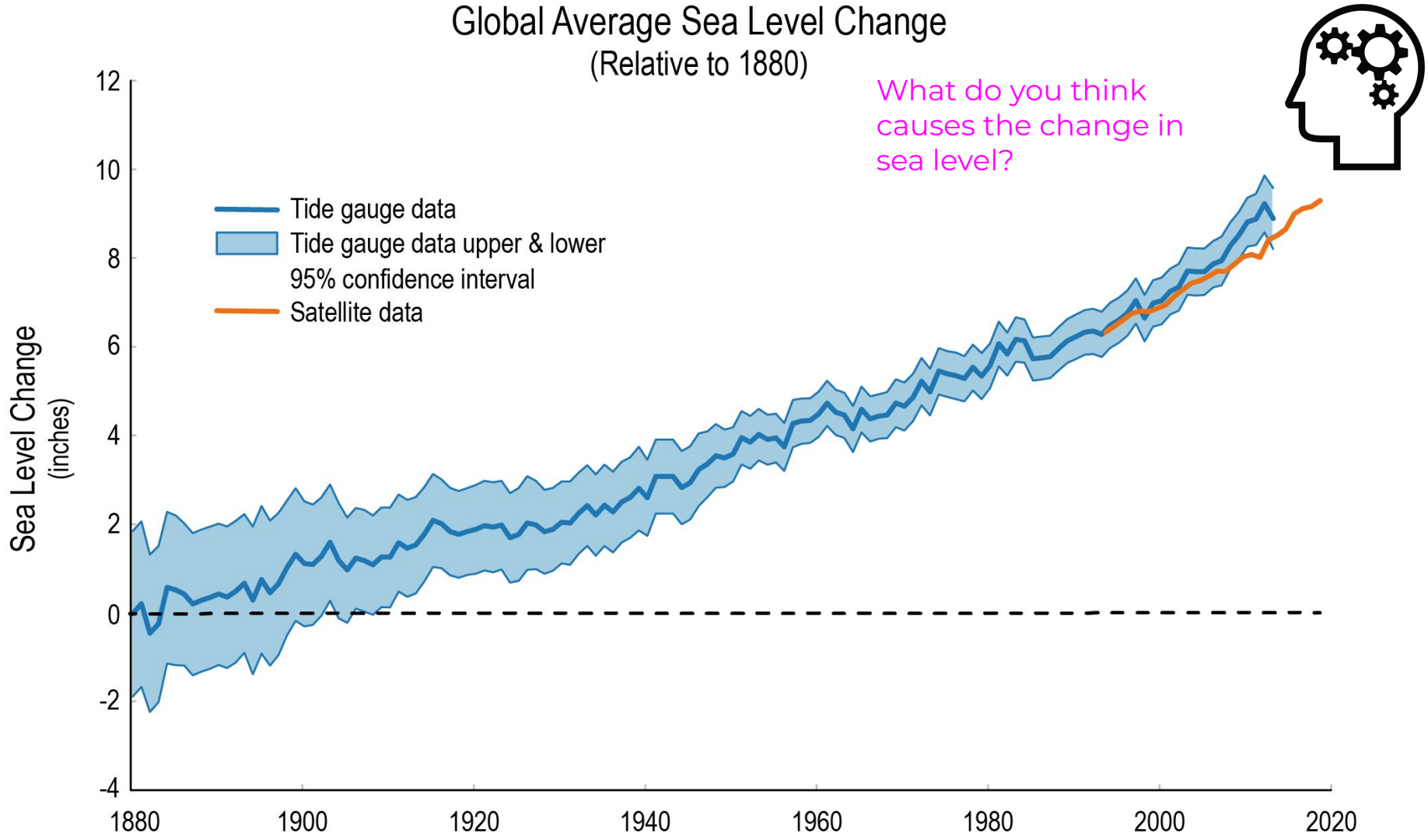
Harvested wood

- When emitted (e.g. burned)
- *does not include for later use



CANADA FOREST CARBON -- Emissions from Canada's managed forest lands. Data from National Inventory Reports 2019 & 2020, with full time series for some tables provided to author by Environment Canada. Black line is net forest flux (table 6-5). Yellow area is HWP (table 10s1). Brown top line is sum of two. CHART by Barry Saxifrage at VisualCarbon.org & NationalObserver.com. May 2020

D4.3.4—Melting of landfast ice and sea ice as examples of polar habitat change

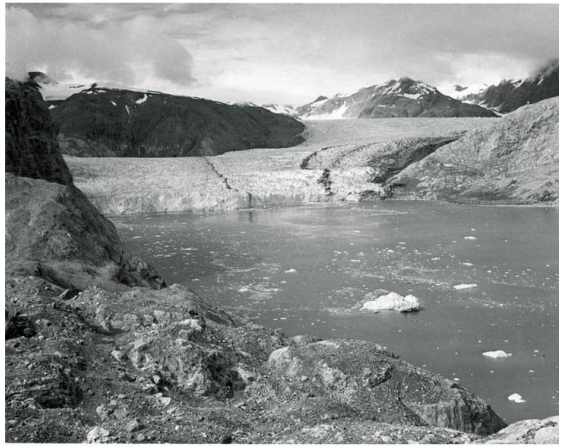

Global Average Sea Level Change
(Relative to 1880)
What do you think causes the change in sea level?
Tide gauge data
Tide gauge data upper & lower 95% confidence interval
Satellite data
Sea Level Change (inches)
12
10
8
6
4
2
0
-2
-4
1880
1900
1920
1940
1960
1980
2000
2020

Sea level rise:

Approx 50% temperature rise, 50% due to glacial melt.



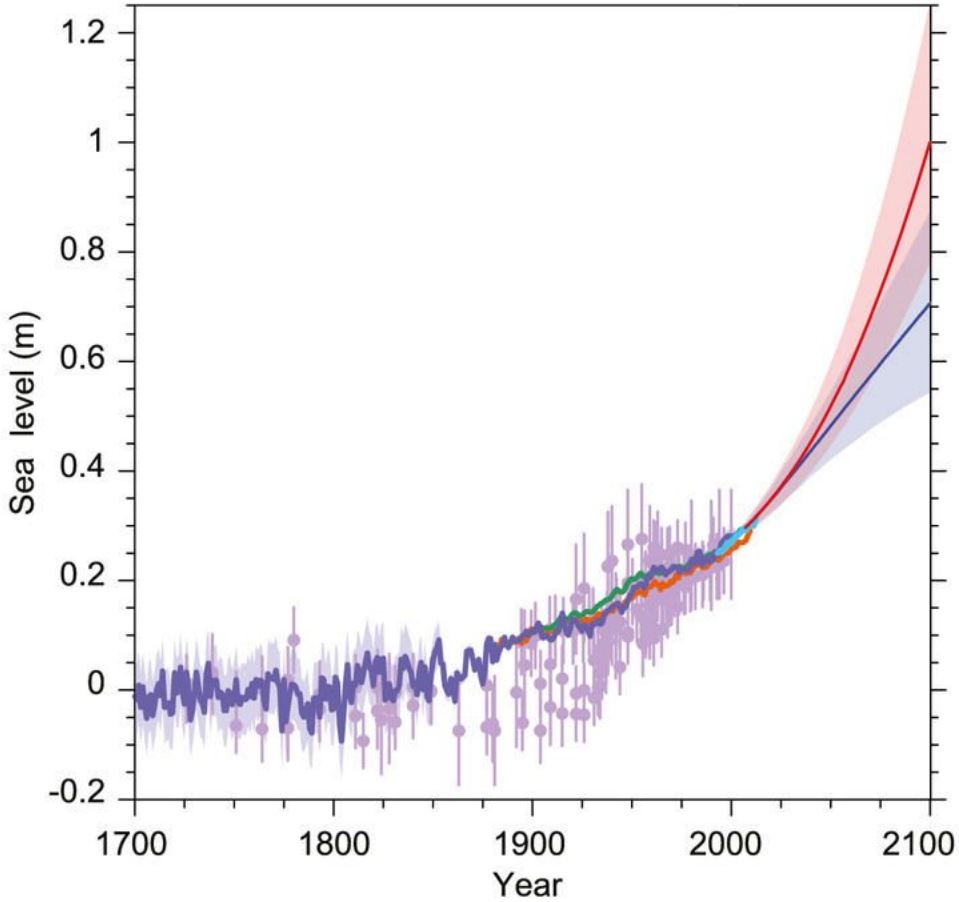
1941
Muir Glacier, Alaska



1950

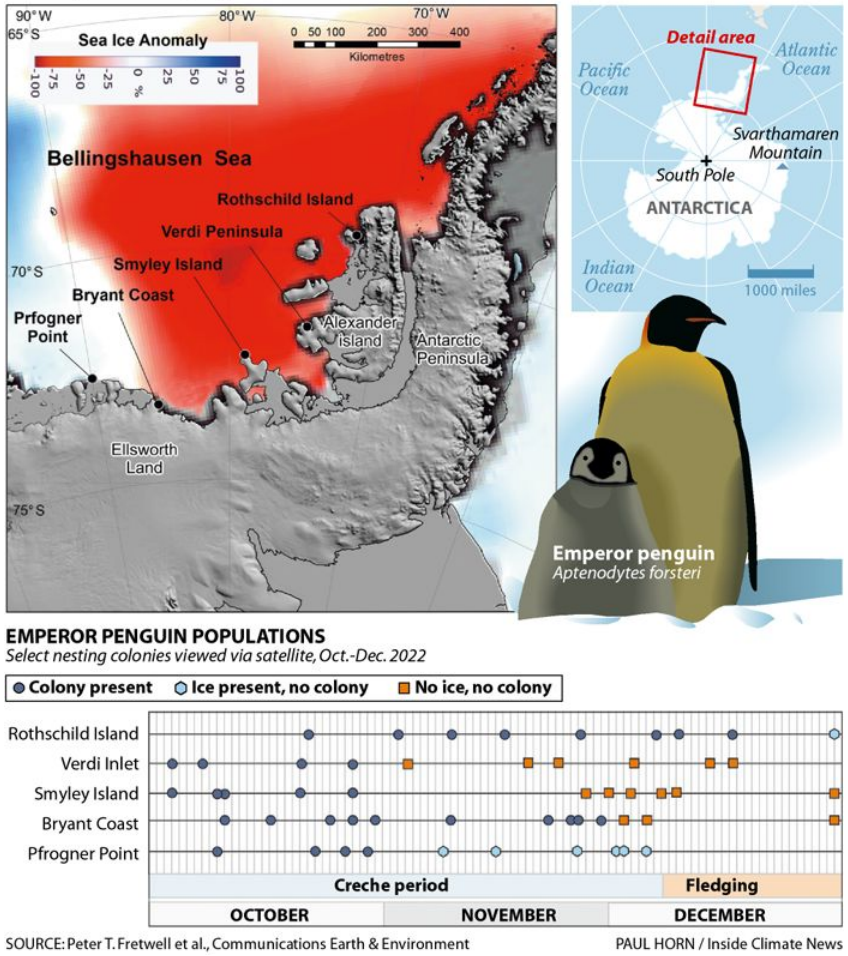


2004

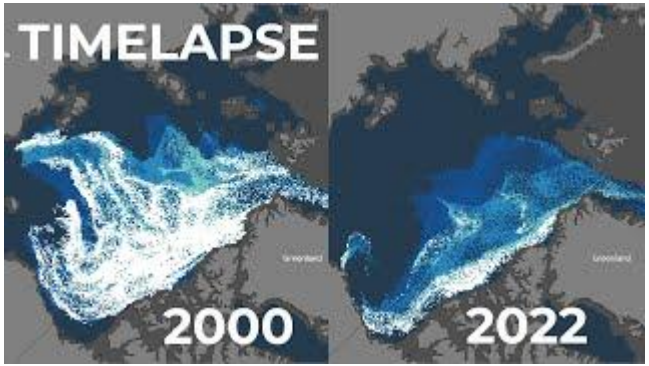


Sea-level rise projections by the IPCC Fifth Assessment Report, Working Group 1 (Figure 13.27). Historical data, altimetry and tide-gauges make up the sea-level rise prior to 2015, while the red line projects business-as-usual emissions (RCP8.5) and the blue line projects strong emission reductions (RCP2.6).

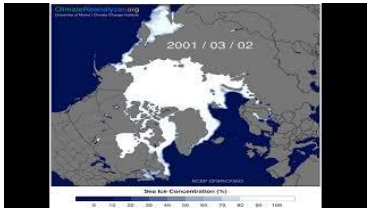
Loss of breeding grounds for the Emperor penguin in the Antarctic



Sea ice habitat for walruses in the Arctic







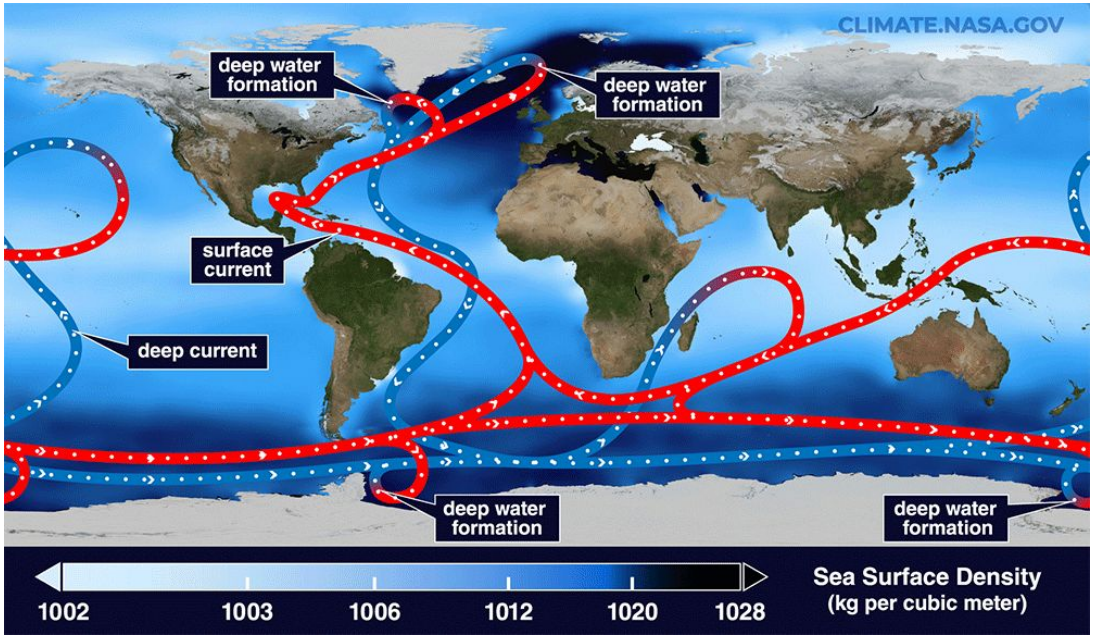
D4.3.5—Changes in ocean currents altering the timing and extent of nutrient upwelling

Generally:

Cold, salty water sinks at the poles

Warmer water rises by equator

Creates currents that mix oxygen and nutrients through oceans.

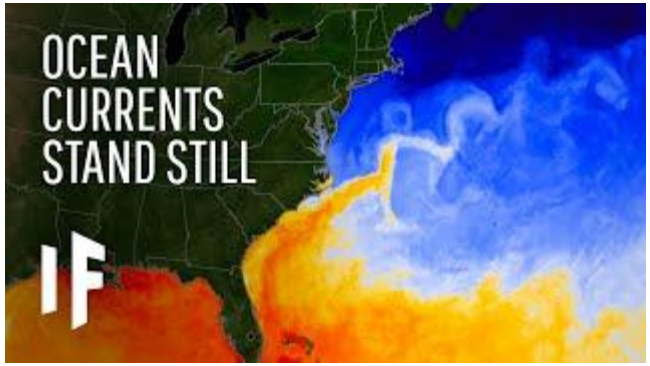


Warmer temperatures at poles

Freshwater ice melts into oceans

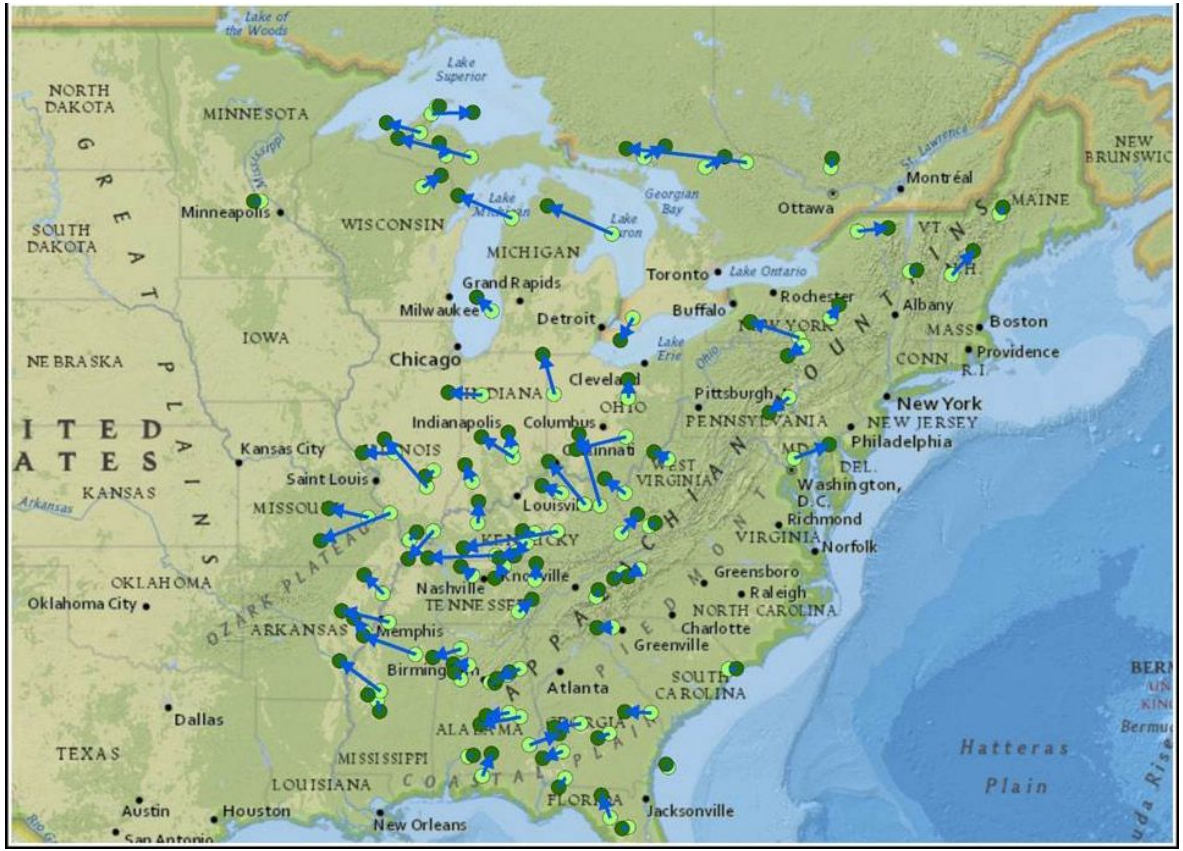
Warmer, less salty water does not sink.

Ocean currents reduced.

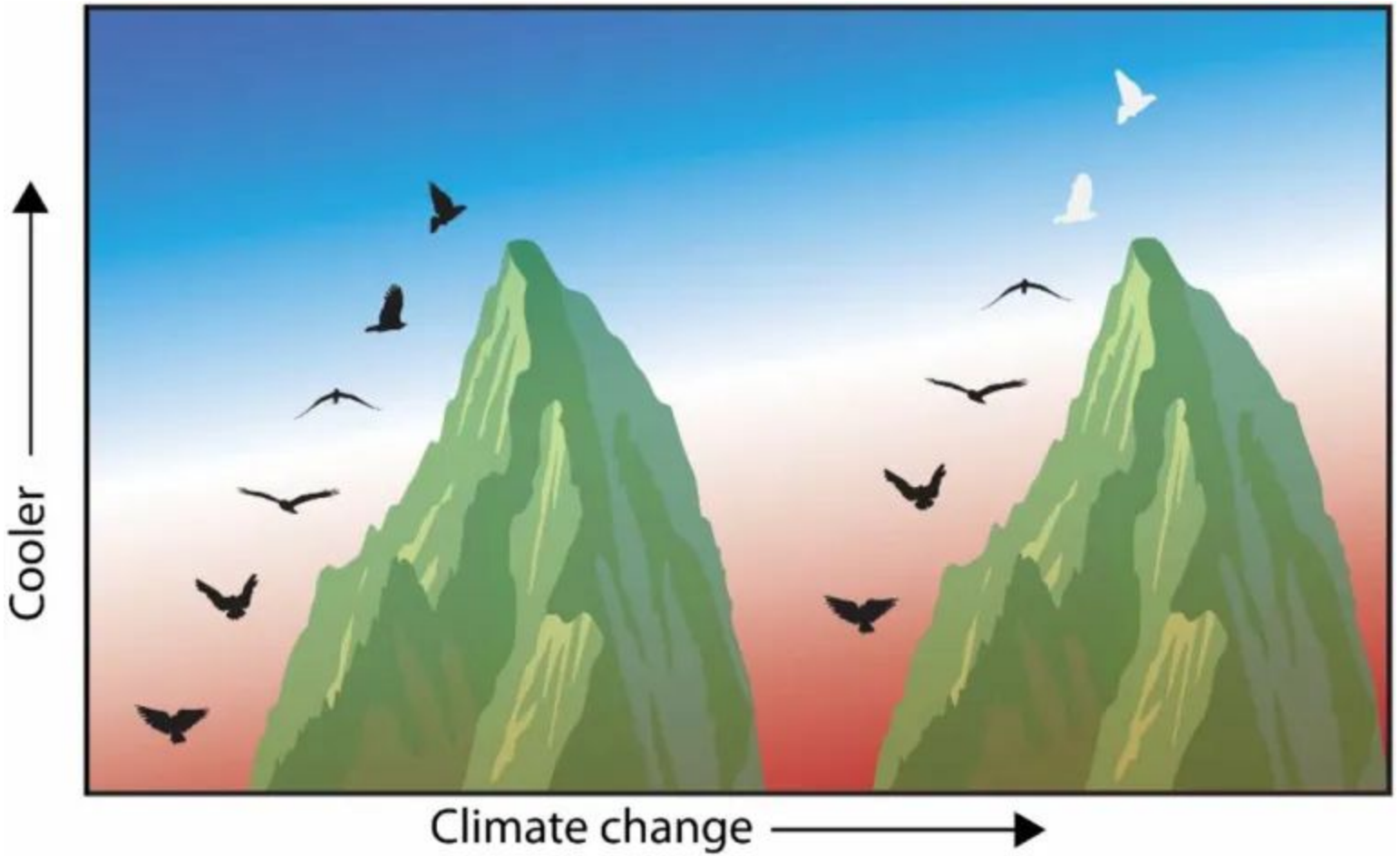

OCEAN
CURRENTS
STAND STILL
IF


CLIMATE CRISIS
GULF STREAM IN PERIL

D4.3.6—Poleward and upslope range shifts of temperate species



A recent study in the Eastern U.S. showed that, over the last three decades, hardwoods ranged west at an average rate of one and a half kilometers each year, while softwoods shifted north an average of one kilometer per year.
Photo Credit: Songlin Fei


Cooler
Climate change

Discuss this graph

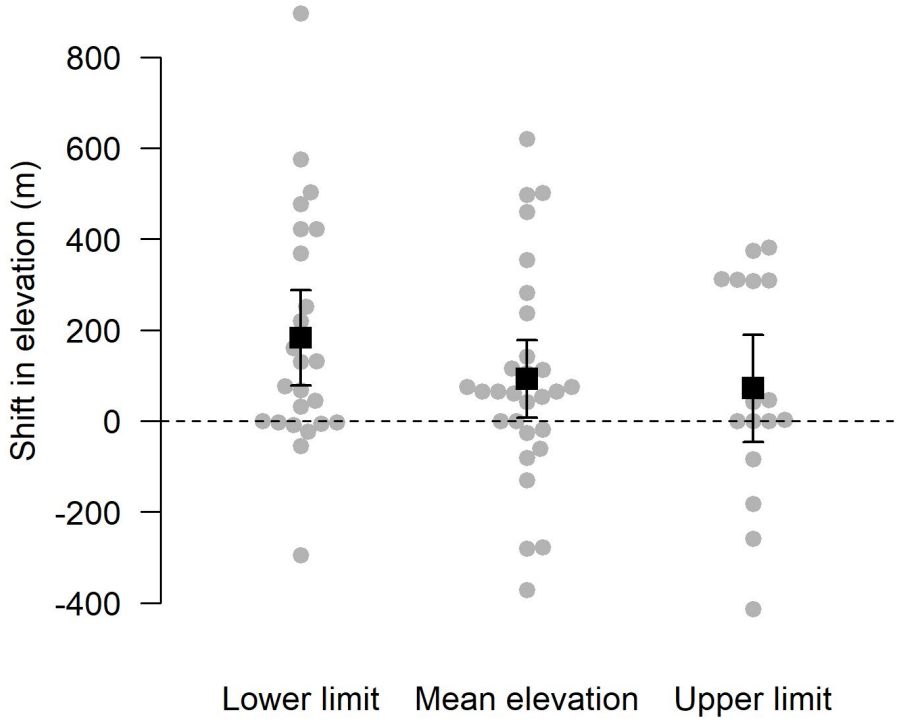


Fig 3. Shifts in mean elevation and lower and upper elevational range limits of 29 species of understory bird species between 1980 and 2019 in the Usambara Mountains, Tanzania.

Elevational shifts were corrected for differences in elevation expected by chance. Each point represents a species. Black squares and bars represent the mean and 95% confidence intervals across species.

https://doi.org/10.1371/journal.pone.0248712.g003

Between 1980 and 2019 understory birds in Tanzania migrated a mean 100m upslope

The lower range moved up more significantly (220m), while lower range moved up less than the mean average.

Birds have a fixed tolerance range

As climate warms they must move to higher altitudes to stay within their tolerance range

As there is a limit to how high they can go their habitat shrinks in size

D4.3.7—Threats to coral reefs as an example of potential ecosystem collapse

Coral reefs

“Rainforests of the sea”

Support incredible biodiversity

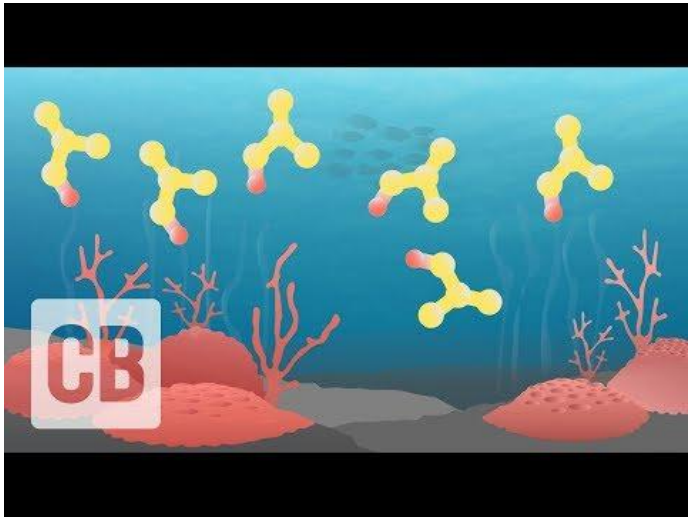
Help with protecting coastlines

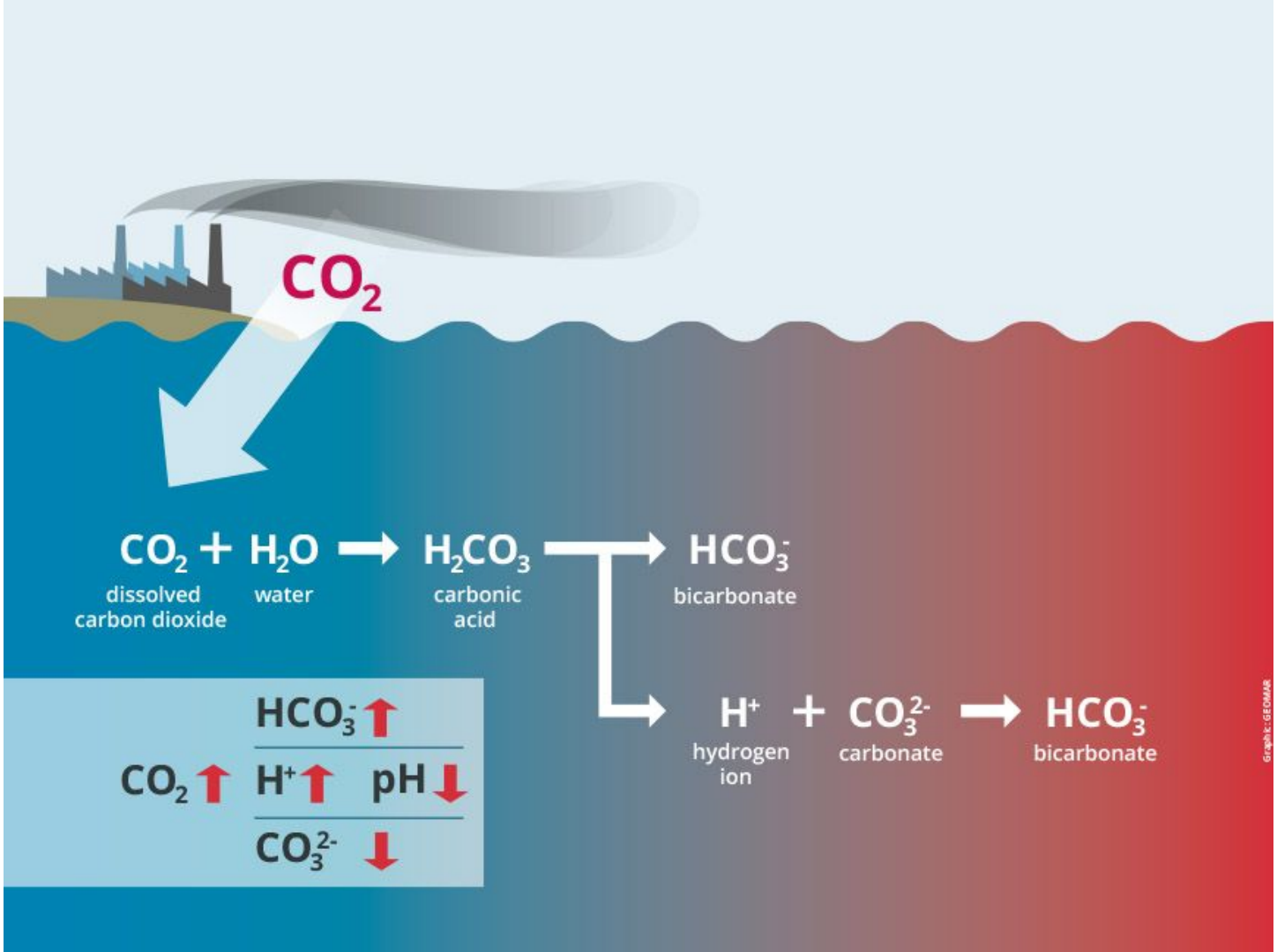
Source of foods and medicines

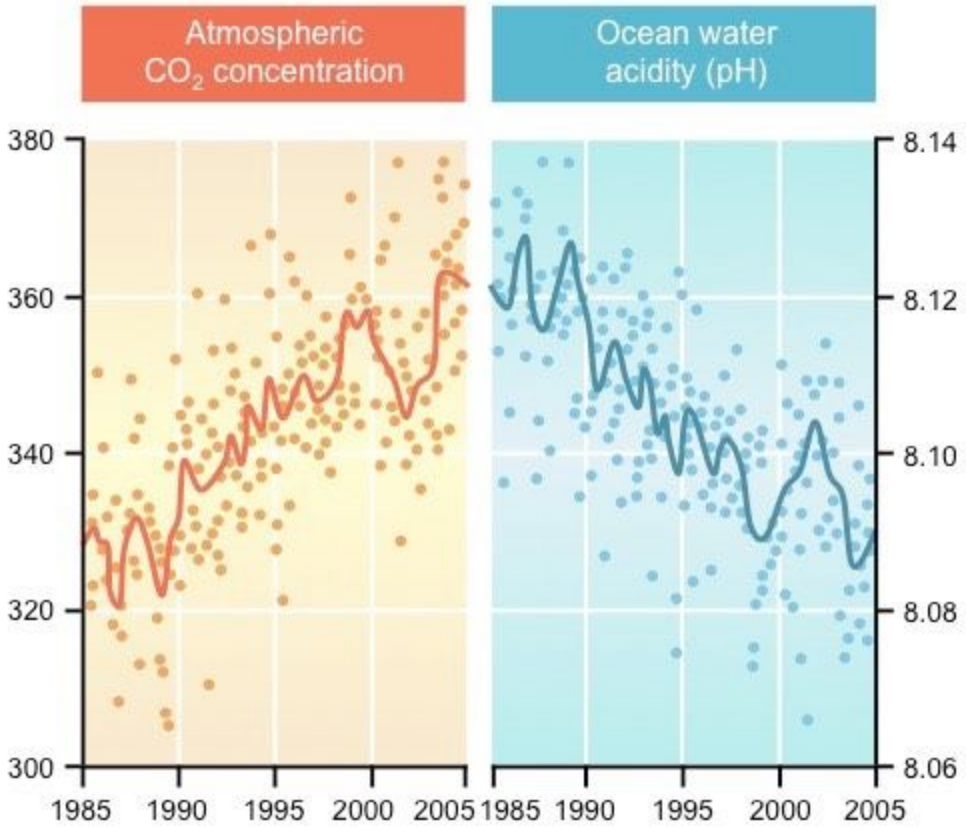


Oceans absorb heat from the atmosphere, but also CO_2

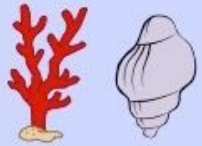
A third of CO_2 emissions absorbed by oceans.




CO_2
CO_2 + H_2O → H_2CO_3 → HCO_3^-
dissolved carbon dioxide
water
carbonic acid
bicarbonate
HCO_3^- ↑
CO_2 ↑ H^+ ↑ pH ↓
CO_3^{2-} ↓
H^+ + CO_3^{2-} → HCO_3^-
hydrogen ion
carbonate
bicarbonate
Graphic: GEOMAR


Atmospheric CO_2 concentration
Ocean water acidity (pH)
380
360
340
320
300
8.14
8.12
8.10
8.08
8.06
1985
1990
1995
2000
2005
1985
1990
1995
2000
2005


Shell deterioration in pH solution of 7.8
Photo credit: David Liittschwager/National Geographic
0 days
15 days
45 days
30 days

Condition	Chemical Reaction	Effect
Normal Levels of Atmospheric CO_2	Carbon Dioxide CO_2 + Water H_2O → Carbonic acid H_2CO_3 → Bicarbonate HCO_3^- Carbonate CO_3^{2-} + H^+ → Bicarbonate HCO_3^- Carbonate CO_3^{2-} + Ca^{2+} → Calcium carbonate $CaCO_3$	Normal pH 8.2 Thick shells and healthy coral 
Elevated Levels of Atmospheric CO_2	Carbon Dioxide CO_2 + Water H_2O → Carbonic acid H_2CO_3 → Bicarbonate HCO_3^- Carbonate CO_3^{2-} + H^+ → Bicarbonate HCO_3^- Carbonate CO_3^{2-} + Ca^{2+} → Calcium carbonate $CaCO_3$	Lower pH 7.8 Thin shells and dead coral 

Coral bleaching

Occurs when corals expel their photosynthetic algae

1 Coral receive their colour from algae called zooxanthellae that live on them in a symbiotic relationship

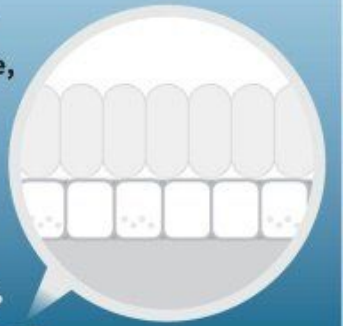
2 Algae use nitrogen, phosphorus and other metabolic waste from coral to generate energy from the sun via photosynthesis



3 Oxygen and other organic products of photosynthesis help coral to grow

Bleaching process

- Under environmental stress, such as a change in temperature, corals expel algae



- Pollution and overexposure to the sun's rays can also cause algae to leave
- Loss of algae causes coral bleaching, making them vulnerable to disease
- Bleached coral will eventually die if they don't regain algae

Source : globalcoralbleaching.org/ NOAA/NationalGeographic/ Oceana.org/teachoceanscience.net

© AFP

D4.3.8—Afforestation, forest regeneration and restoration of peat-forming wetlands as approaches to carbon sequestration



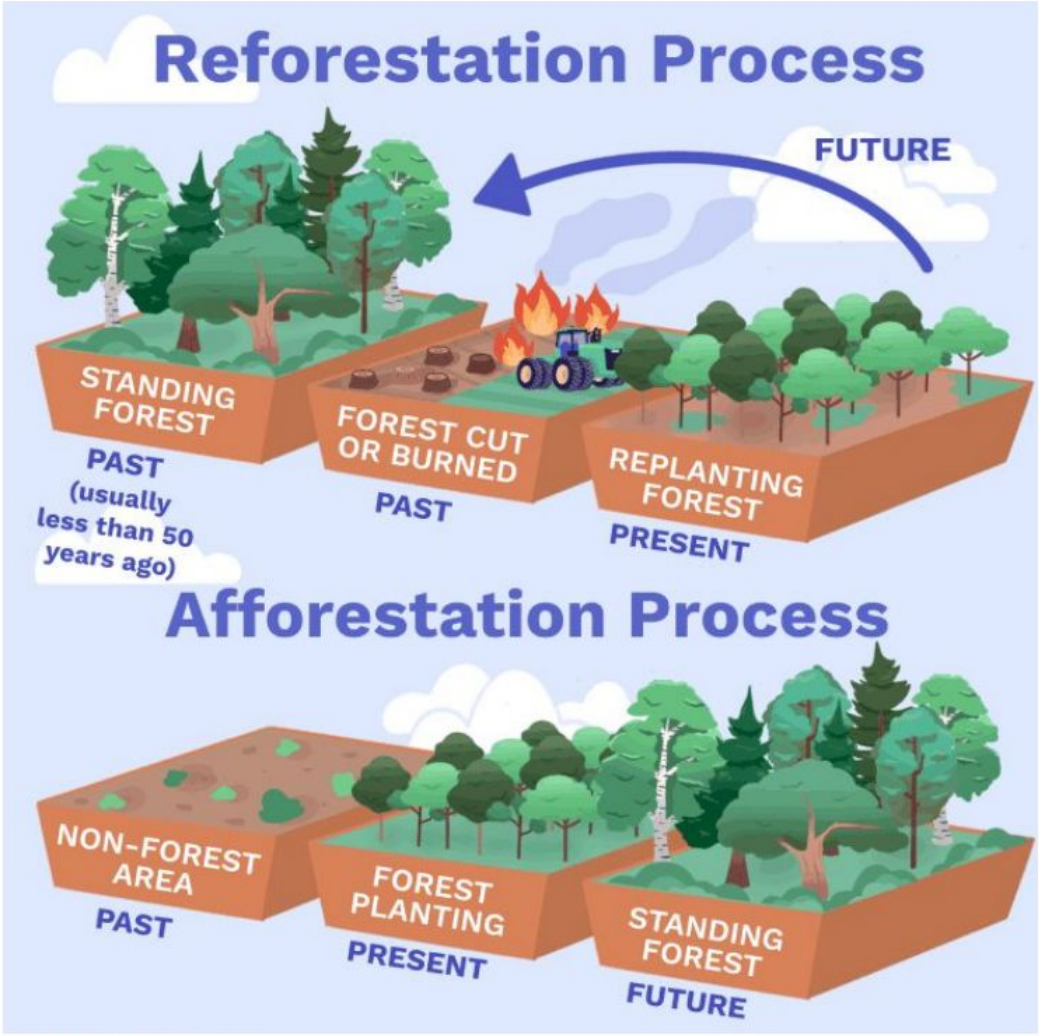
What machine can...

Remove carbon from the atmosphere,

Increase quality of air,

Increase biodiversity,

And provide a beautiful landscape?



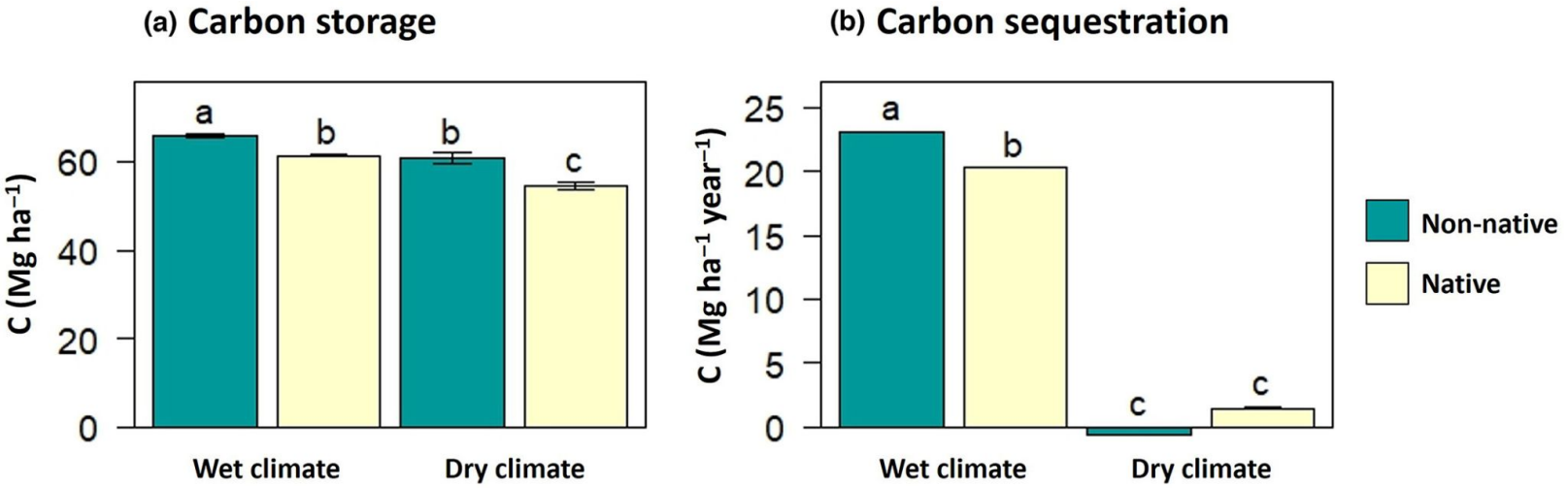
climatescience.org

Nature of Science

There is active scientific debate over whether plantations of non-native tree species or rewilding with native species offer the best approach to carbon sequestration. Peat formation naturally occurs in waterlogged soils in temperate and boreal zones and also very rapidly in some tropical ecosystems.

Sequestration = photosynthesis
Storage = growth

Discuss this graph



Predicted means of carbon storage and sequestration for non-native and native forests in wet and dry climates. Different letters indicate significant differences between categories after accounting for multiple-comparison Bonferroni correction. Error bars represent standard errors.

Native vs Non Native Trees

Native Trees

Creates Beauty and are Low Maintenance

Native plants generally require little maintenance. offer beautiful flowers, produce abundant colorful fruits and seeds and brilliant seasonal changes in colors from the pale, thin greens of early spring, to the vibrant yellows and reds of autumn.

Creates Healthy Places for People

By choosing native plants for your landscaping, you are creating a healthier place for yourself, your family, and your community.

Helps the Climate

Native trees help fight climate change. Through the natural process of photosynthesis, trees absorb CO_2 and other pollutants particulates, then store the carbon and emit pure oxygen.

Conserves Water

It protects groundwater quality and helps maintain high water levels. Root systems take up nitrogen and phosphorus.

Attracts Biodiversity

In addition to providing vital habitat for birds, many other species of wildlife benefits as well. The colorful array of butterflies and moths, including the iconic monarch, the swallowtails, tortoiseshells, and beautiful blues, are all dependent on very specific native plant species.

Non Native Trees

High Maintenance

The non-native tree is not always available they require more water and more work keeping them alive.

Do Not Attract Biodiversity

They do not attract butterflies and wildlife in the area of non native trees.

Change Ecosystems

Change ecosystems and habitats and have non-biotic effects, such as reducing or impeding water flow leading to flooding, or changing the pH or the chemical composition of the soil, or lock up nutrients.

Less Environmental Benefit

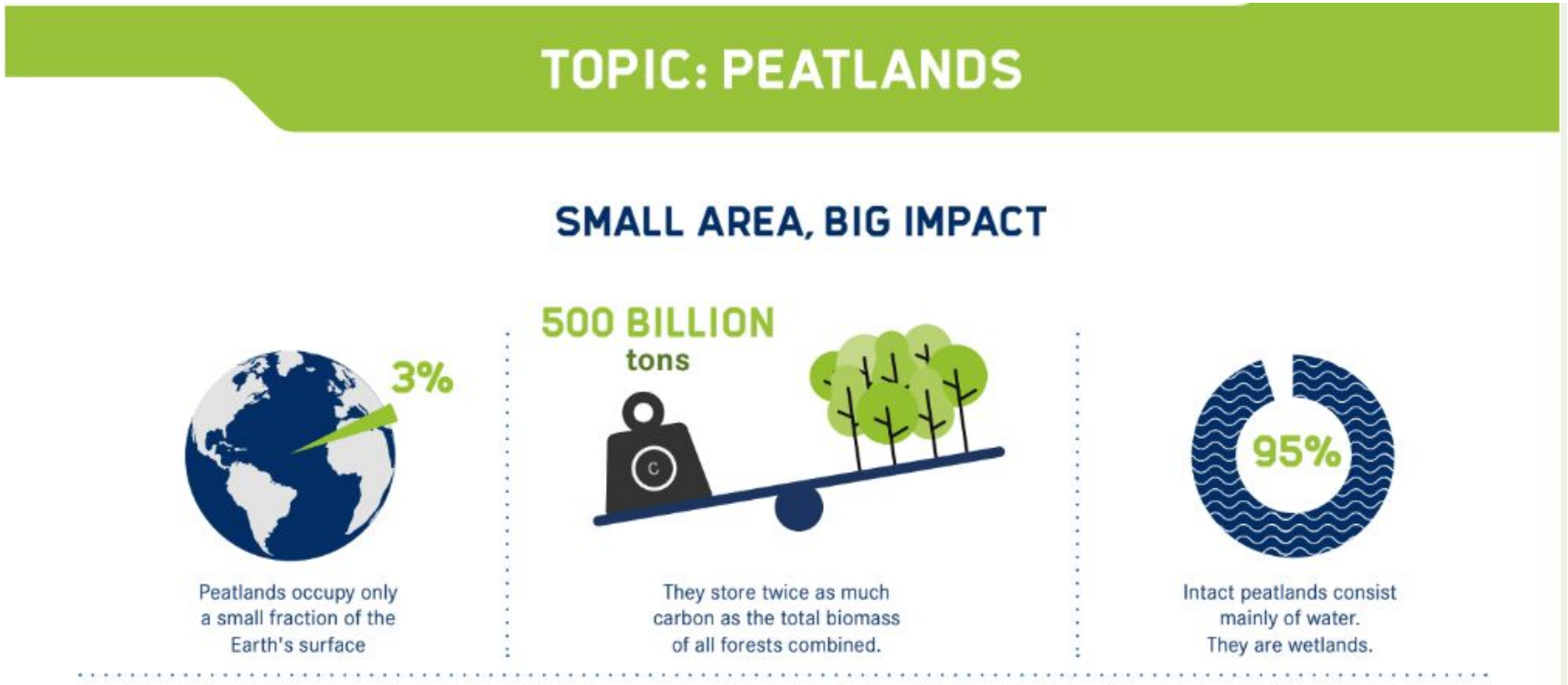
The non-native plants tend to be a bit worse for the local environments. Just as some are more resistant to local diseases, so too are some more prone to them or bring disease that affects native plants.

Affect Native Plants

Just as some are more resistant to local diseases, so too are some more prone to them or bring disease that affects native plants.



Rewetting drained peatlands

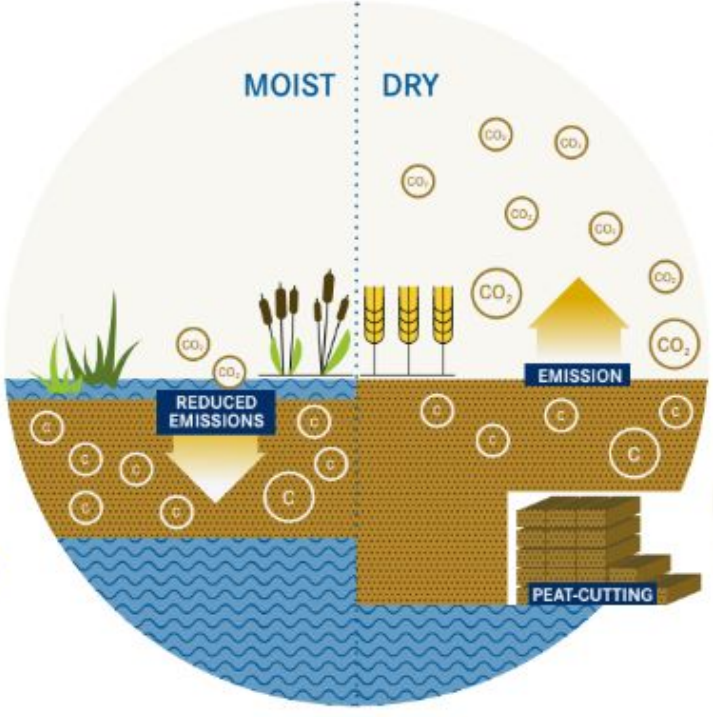


HOW PEATLANDS AFFECT THE CLIMATE

Natural peatlands protect the climate.
Wet peatlands store CO_2 from the atmosphere in the form of carbon (C), because the remains of animals, plants and organisms do not completely decompose in the moist soil.

The result is peat, a sediment rich in carbon (C). Peat was long used as fuel and as an additive in planting soil.

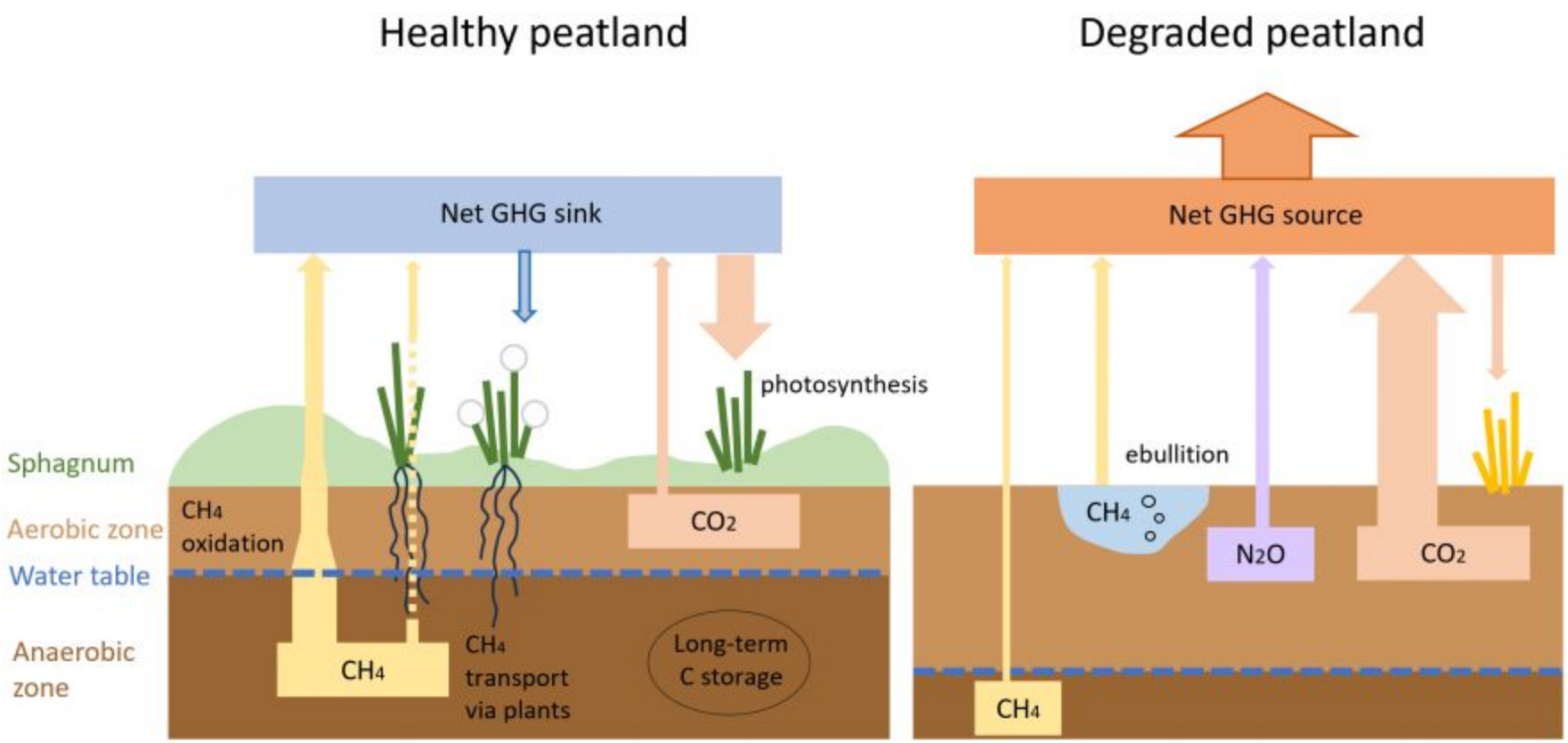
Almost 30% of the carbon stored in the ground is in peatlands.



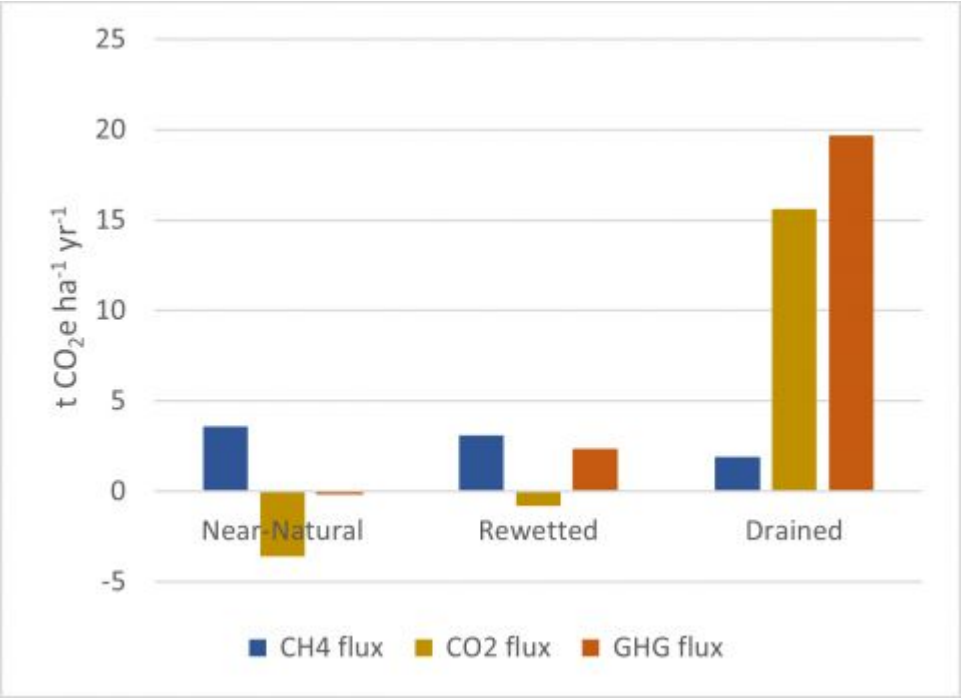
98 % of the peatlands in Germany have been drained.

The drainage of peatlands intensifies climate change.
Many peatlands have been drained for agriculture and peat-cutting. This allows oxygen to reach the organic material full of carbon (C).

It is decomposed by microbes that produce CO_2. The stored carbon turns into a greenhouse gas that heats up the atmosphere.


Healthy peatland
Degraded peatland
Net GHG sink
Net GHG source
photosynthesis
Sphagnum
Aerobic zone
Water table
Anaerobic zone
CH4 oxidation
CO2
CH4
CH4 transport via plants
Long-term C storage
ebullition
CH4
N2O
CO2
CH4

Discuss this graph



Methane short lived 12 years vs CO_2 which lasts 100s of years

Offset by amount of carbon stored.


ora
www.ora.tv



Arguments against climate change:

1. Climate change denier claim: “This is the coldest winter we’ve had in years! So much for global warming.”

2. “Climate change is natural and normal—it’s happened at other points in history.”

3. “There’s no consensus among scientists that climate change is real.”

4. “Plants and animals can adapt.”

5. “Climate change is good for us.”

How would you respond?

Activity

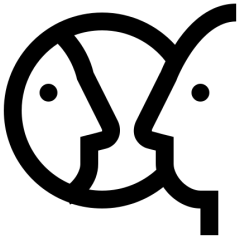
How would you convince a skeptic that climate change is real?

Individually, create an A2 poster outlining:

- What are greenhouse gasses
- What is global warming?
- How does CO2 relate to climate change?
- **What is the evidence?**
- What can we do?


SORRY

Can you...



Explain the anthropogenic causes of climate change?

Describe some positive feedback loops in global warming?

Describe the impact of melting polar ice?

Describe how climate change may shift the range of habitats?

Describe the effects of climate change on coral reefs?

Discuss some methods of carbon sequestration?